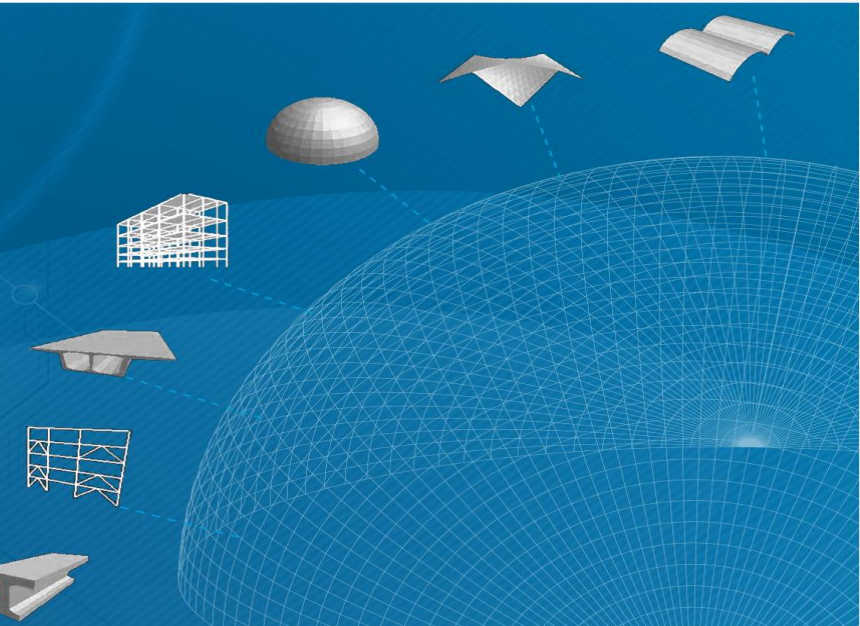
CE 412 Design of Concrete Structures

Lecture 5:

Member Behavior in Combined Axial Load and Flexure – Long Columns



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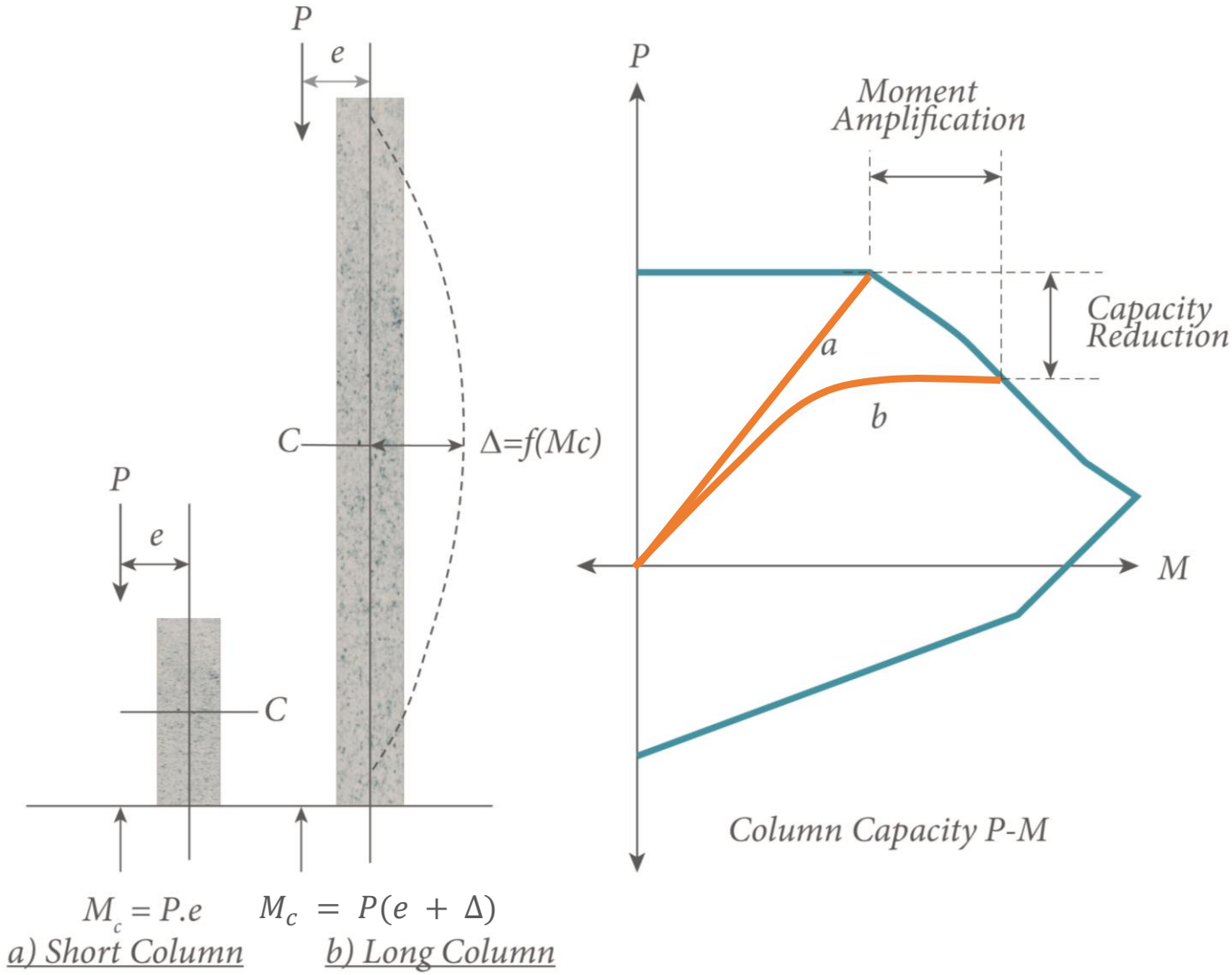
Member Capacity vs. Section Capacity

Long Columns

Member Capacity

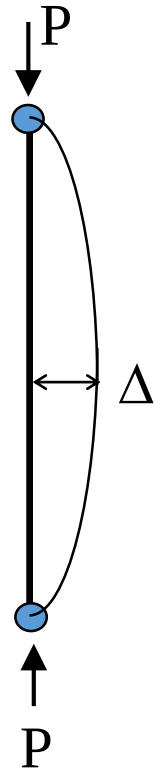
- The member capacity is based on the capacity of cross-section at various locations along the member length.
- The member capacity is almost always less than cross-section capacity at critical location.
- The reduction in member capacity is due to the stability considerations, P-Delta effects and nonlinearity in member behavior, effect of boundary conditions and interaction with other load configurations etc.

What is Slenderness Effect?



Main Issues in Column Slenderness

- Overall Objective
  - To estimate magnification of the “elastic actions” due to geometric and material in- elasticity or nonlinearity.
- Real Situation
  - Geometric Effect Alone $M = M_0 + P\Delta$
  - Material Effect Alone
    - Δ_o based on $E_0 I_g$ Cracking $I_g \rightarrow I_{eff}$
    - Δ based on $(EI)_{modified}$ (Nonlinear E_c)
- Correct Approach
  - Nonlinear analysis that includes effect of geometric and material nonlinearity of “entire” structure.
- Approximate Approach
  - Moment magnification factor $M = \delta Mo$

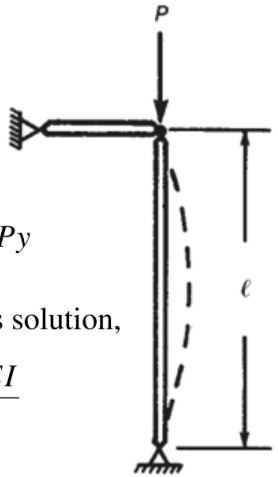


Buckling of Axially Loaded Elastic Columns

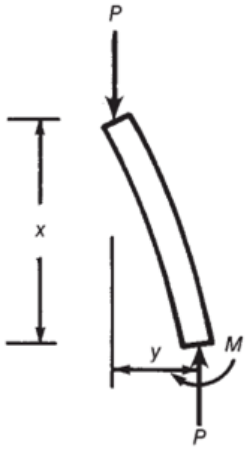


(c) Number of half-sine waves.

Buckling of a pin-ended column



(a) Column.



(b) Free-body diagram.

The differential equation for this column is

$$EI\frac{d^2y}{dx^2} = -Py$$

In 1744, Leonhard Euler derived Eq. (12-3) and its solution,

$$P_c = \frac{n^2\pi^2EI}{\ell^2}$$

where

EI = flexural rigidity of column cross section
ℓ = length of the column
n = number of half-sine waves in the deformed shape of the column

Cases with $n = 1$, 2, and 3 are illustrated in Fig. 12-4c. The lowest value of P_c will occur with $n = 1.0$. This gives what is referred to as the *Euler buckling load*:

$$P_E = \frac{\pi^2EI}{\ell^2} \qquad (12\text{-}5)$$

Such a column is shown in Fig. 12-5a. If this column were unable to move sideways at midheight, as shown in Fig. 12-5b, it would buckle with $n = 2$, and the buckling load would be

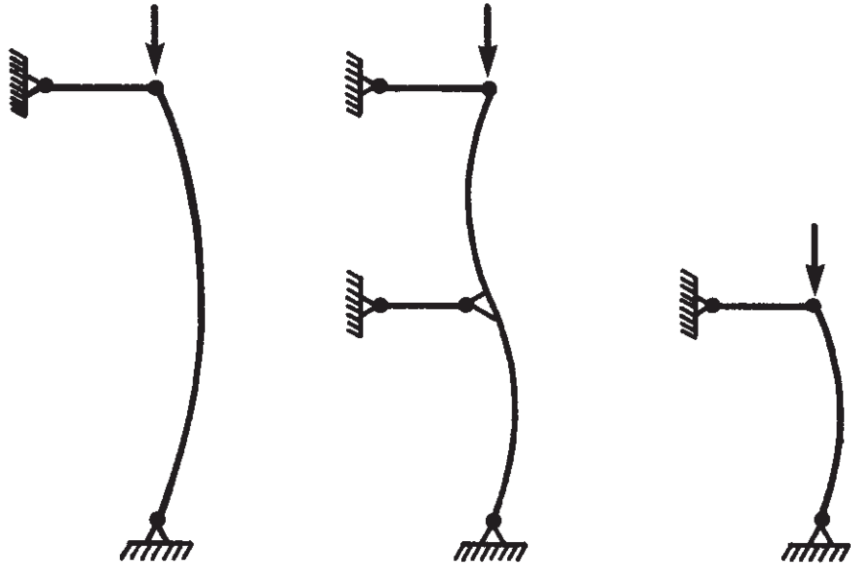
$$P_c = \frac{2^2\pi^2EI}{\ell^2}$$

which is four times the critical load of the same column without the midheight brace.

The Concept of Effective Length Factor, K

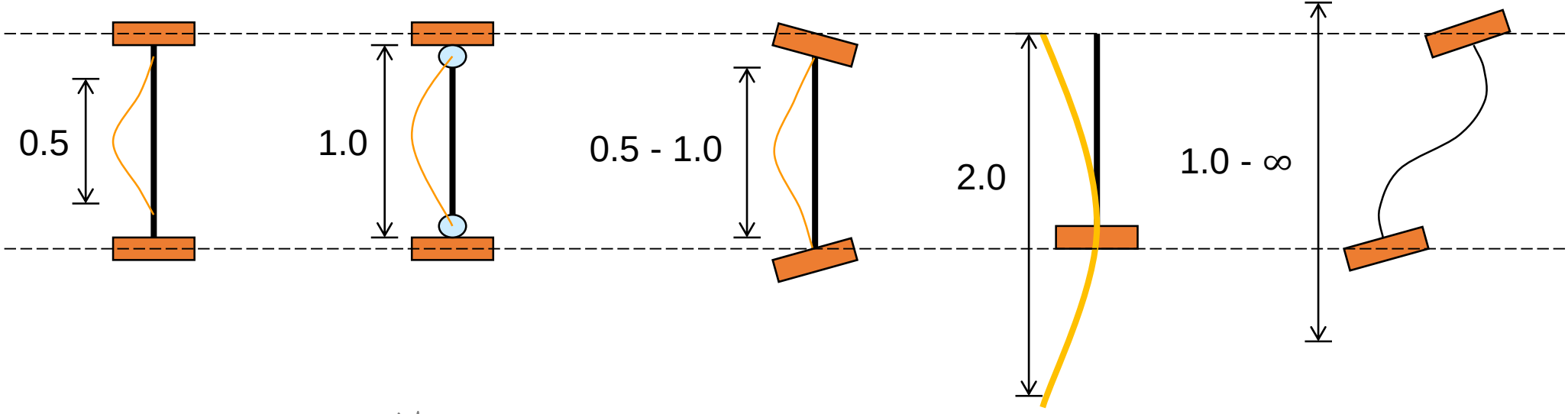
Another way of looking at this involves the concept of the *effective length* of the column. The effective length is the length of a pin-ended column having the same buckling load. Thus the column in Fig. 12-5c has the same buckling load as that in Fig. 12-5b. The effective length of the column is $\ell/2$ in this case, where $\ell/2$ is the length of each of the half-sine waves in the deflected shape of the column in Fig. 12-5b. The effective length, $k\ell$, is equal to ℓ/n. The *effective length factor* is $k = 1/n$. Equation (12-4) is generally written as

$$P_c = \frac{\pi^2 EI}{(k\ell)^2} \tag{12-6}$$



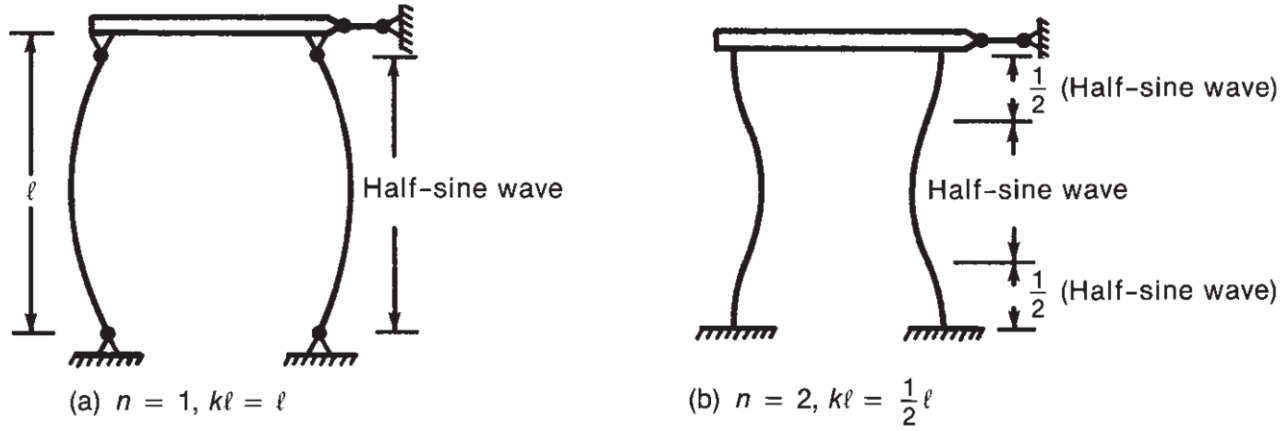
Effective Length Factor, K

- To account for “Axial-Flexural Buckling”.
- Indicates the “total bent” length of column between inflection points.
- Can vary from 0.5 to Infinity. Most common range is 0.75 to 2.0.



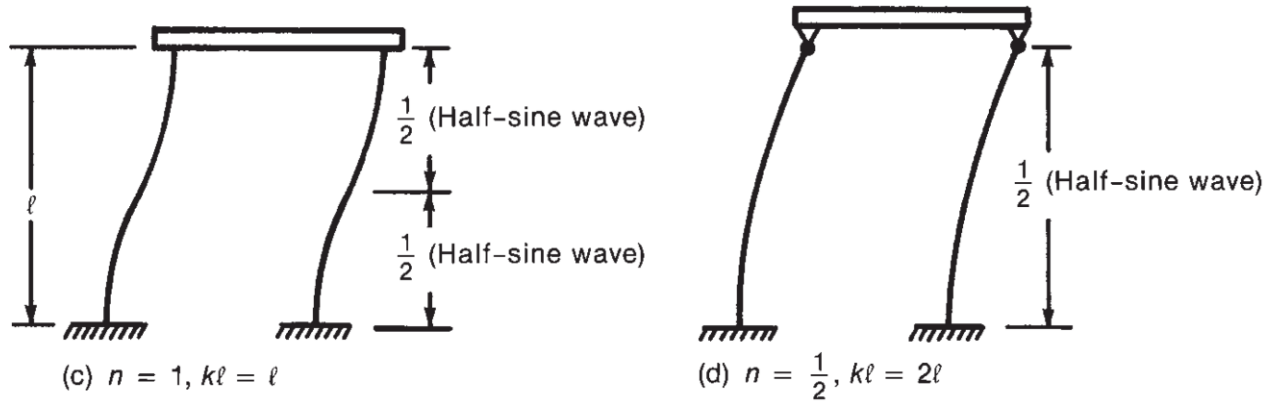
Effective lengths of idealized columns

- Frames a and b are prevented against deflecting laterally. They are said to be braced against sidesway.
- Frames c and d are free to sway laterally when they buckle. They are called unbraced or sway frames.
- The critical loads of the columns are in the ratio 1:4:1:1/4



(a) $n = 1, k\ell = \ell$ (b) $n = 2, k\ell = \frac{1}{2}\ell$

Frames braced against sidesway.



(c) $n = 1, k\ell = \ell$ (d) $n = \frac{1}{2}, k\ell = 2\ell$

Frames free to sway laterally.

Determination of K

Unbraced Frames	$K = \frac{20 - G_m}{20}\sqrt{1 + G_m} \quad for\ G_m < 2$ $K = 0.9\sqrt{(1 + G_m)} \quad for\ G_m \geq 2$
Braced Frames (smaller of)	$K = 0.7 + 0.05(G_T + G_B) \leq 1.0$ $K = 0.85 + 0.05 G_m \leq 1.0$

$$G = \frac{\sum(EI / L_C) \quad Columns}{\sum(EI / L) \quad Beams}$$

$K\ \alpha\ G$ $\quad$ G *Increase,* K *Increases*

$G_T = Top\ End$

$G_B = Bottom\ End$

$G_m = Minimum\ of\ G_T\ and\ G_B$

For members which are a part of framed structure

Determination of Stiffness EI (ACI)

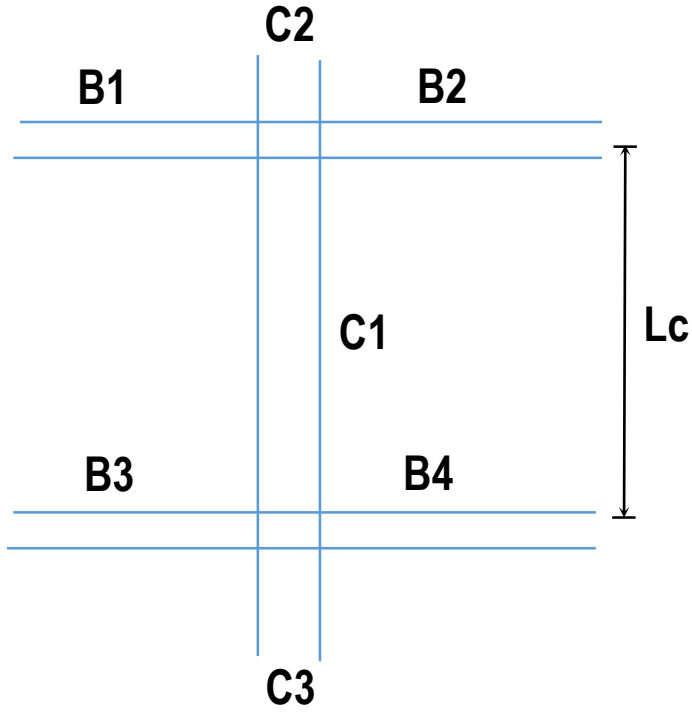
$$EI = \frac{0.2E_C I_g + E_s I_{se}}{1+\beta_d}$$

$$or \quad = \frac{0.4E_C I_g}{1+\beta_d}$$

- Attempt to include,
  - Cracking, Variable E, Creep effect
  - Geometric and material non-linearity
- I_g = Gross Moment of Inertia
- I_{se} = Moment of Inertia of rebars
- β_d = Effect of creep for sustained loads

More about Factor K

- How about “I” Gross? Cracked? Effective?
- ACI Rules Beams I = 0.35 Ig, Column I = 0.7Ig



- E for column and beams may be different

$$Example = G = \frac{E(I_{C1} + I_{C2})}{E(I_{B1} + I_{B2})}$$

Factors Effecting Slenderness Effect

- “Effective” Length
  - Length used for moment integration
  - End Framing and Boundary Conditions
  - Lateral Bracing Conditions
- “Effective” Stiffness
  - Cross-sections Dimensions and Proportions
  - Reinforcement amount and Distribution
  - Modulus of Elasticity of Concrete and Steel
  - Creep and Sustained Loads
- Loads
  - Axial Load
  - End Moments and Moments along the Length

Column Design - Slenderness

- What is Slenderness ?
  - When the Buckling Load controls the ultimate capacity.
  - or secondary moments become significant.
- ACI Definition of Slenderness
  - Braced Frames (against side-sway)
    - $Kl/r \leq 32 - 12\,(M1/M2) \;<\; 40$
  - Un-braced Frames (against side-sway)
    - $Kl/r \;\leq\; 22$

Slenderness effects are permitted to be neglected in these cases.

Slender Columns in Structures

The left corner column has a height of 50 times its least thickness.

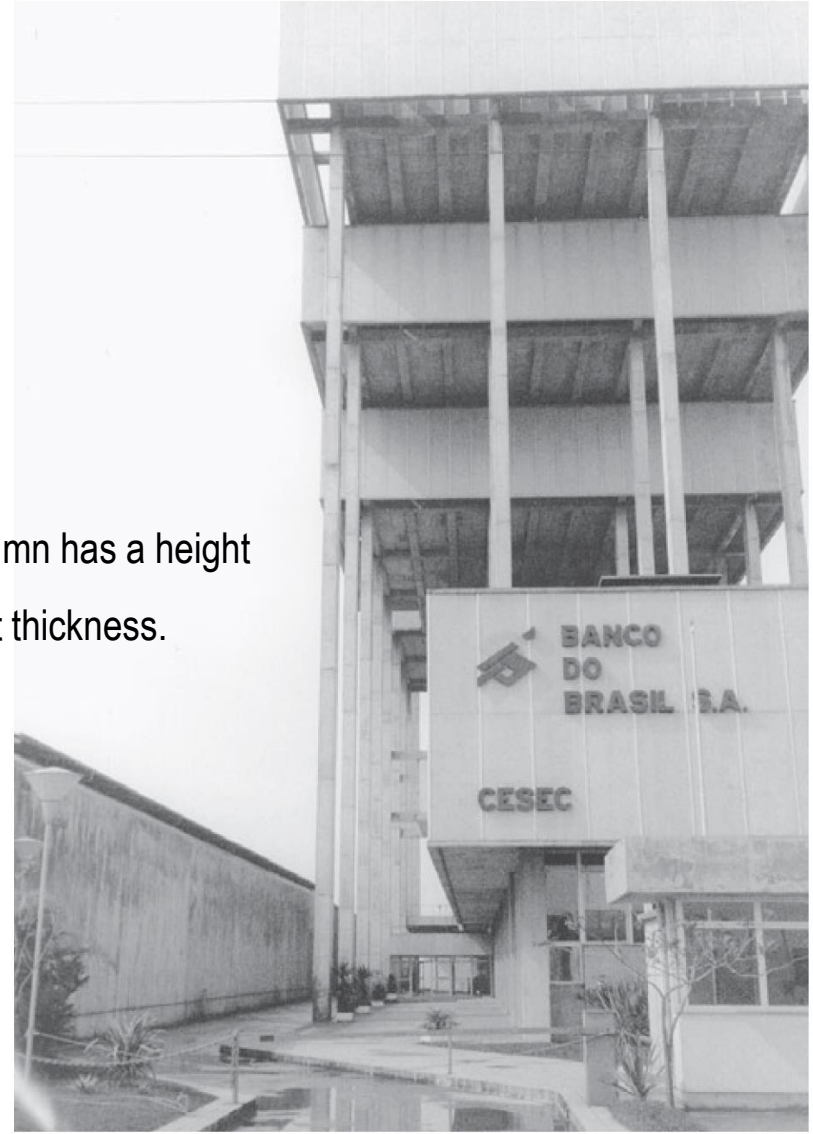


Fig. 12-7
Bank of Brazil building, Porto Alegre, Brazil. Each floor extends out over the floor below it. (Photograph courtesy of J. G. MacGregor.)

Slender Columns in Structures

- Pin-ended columns are rare in cast-in-place concrete construction but do occur in precast construction.
- Most concrete building structures are braced (nonsway) frames, with the bracing provided by shear walls, stairwells, or elevator shafts that are considerably stiffer than the columns themselves.
- Occasionally, unbraced frames are encountered near the tops of tall buildings, where the stiff elevator core may be discontinued before the top of the building, or in industrial buildings where an open bay exists to accommodate a travelling crane.
- Most building columns fall in the short-column category. Exceptions occur in industrial buildings and in buildings that have a high first-floor story for architectural or functional reasons.
- Some bridge piers and the decks of cable-stayed bridges fall into the slender-column category.

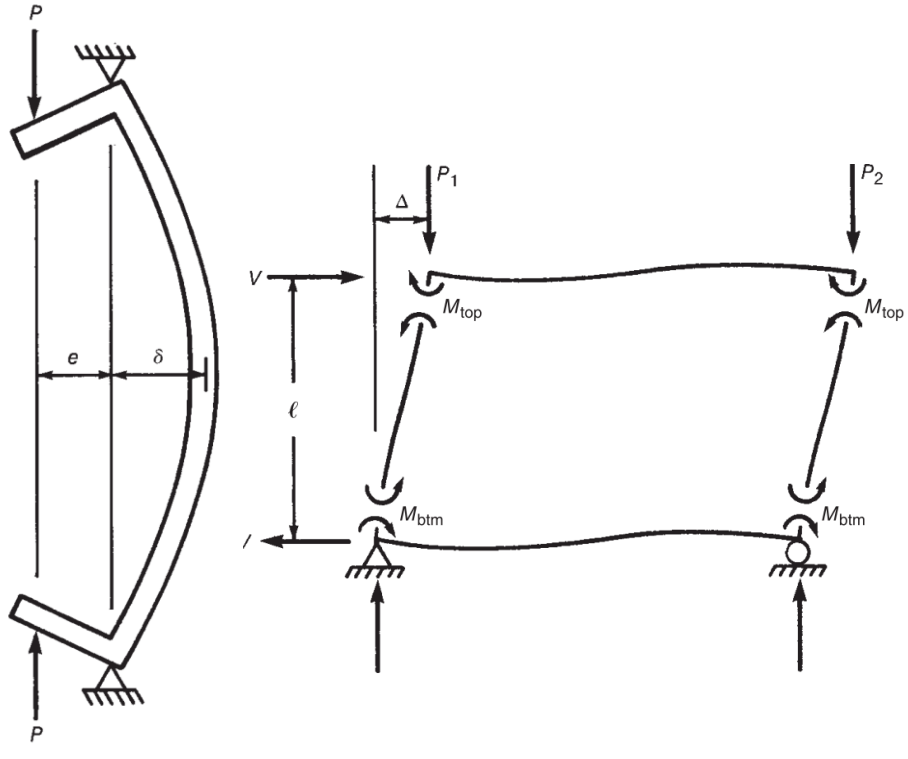
Design of Long Columns

- Behavior, Analysis and Design of Pin-ended Columns
- ACI 318 Moment Magnification Method
- Design of Pin-ended Columns in Non-sway Frames
- Design of Restrained Columns in Non-sway Frames
- Design of Restrained Columns in Sway Frames

Behavior and Analysis of Pin-ended Columns

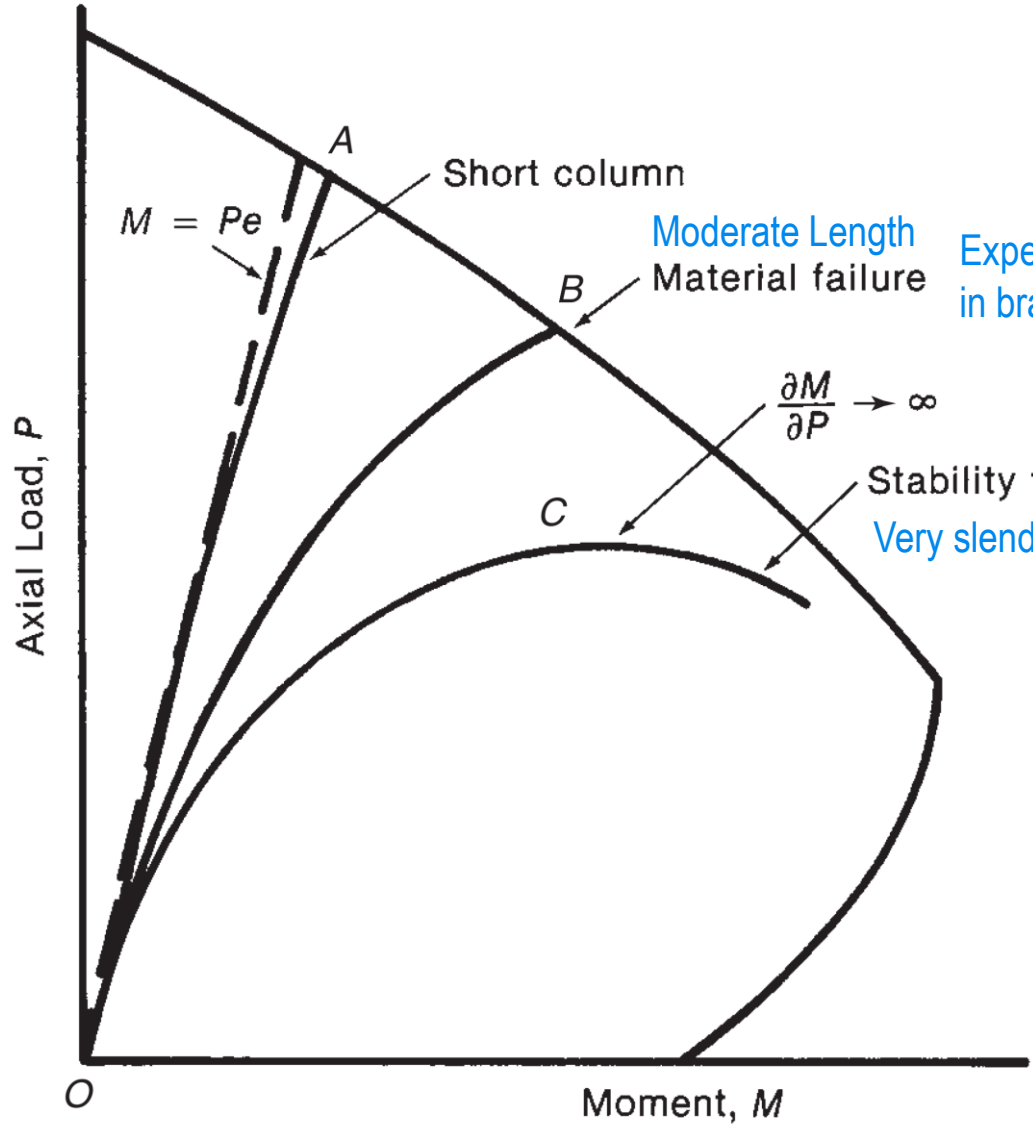
P-δ Moments and P-Δ Moments

- Two different types of second-order moments act on the columns in a frame:
  - P-δ moments. These result from deflections δ, of the axis of the bent column away from the chord joining the ends of the column. The slenderness effects in pin-ended columns and in non-sway frames result from P-δ effects.
  - P-Δ moments. These result from lateral deflections Δ, of the beam–column joints from their original undeflected locations. The slenderness effects in sway frames result from P-Δ moments.



P-δ Moments P-Δ Moments

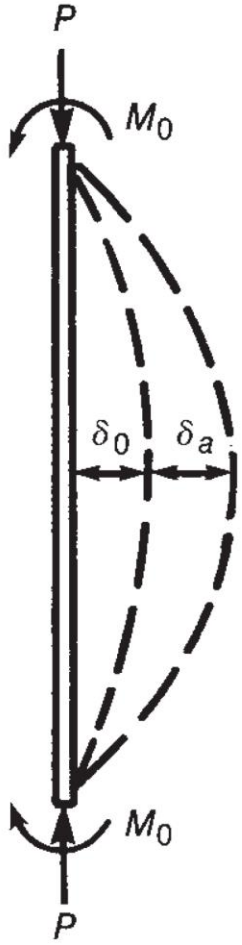
Material Failures and Stability Failures



Assignment

Moment Magnifier for Symmetrically Loaded Pin-Ended Column

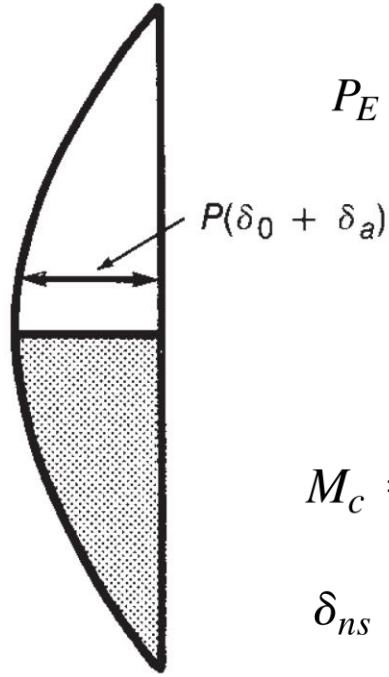
The second-order deflection δ increases as P/P_E increases, reaching infinity when $P = P_E$. The maximum bending moment is $Mc = M_o + P\,\delta$



(a) Deflected column.



(b) M_0.



(c) P–δ Moments.

$$M_c = M_o + P\delta$$

$$\delta = \frac{\delta_o}{1 - P/P_E}$$

$$P_E = \frac{\pi^2 EI}{\ell^2}$$

$$M_c = \delta_{ns} M_o$$

$$\delta_{ns} = \frac{1}{1 - P/P_c}$$

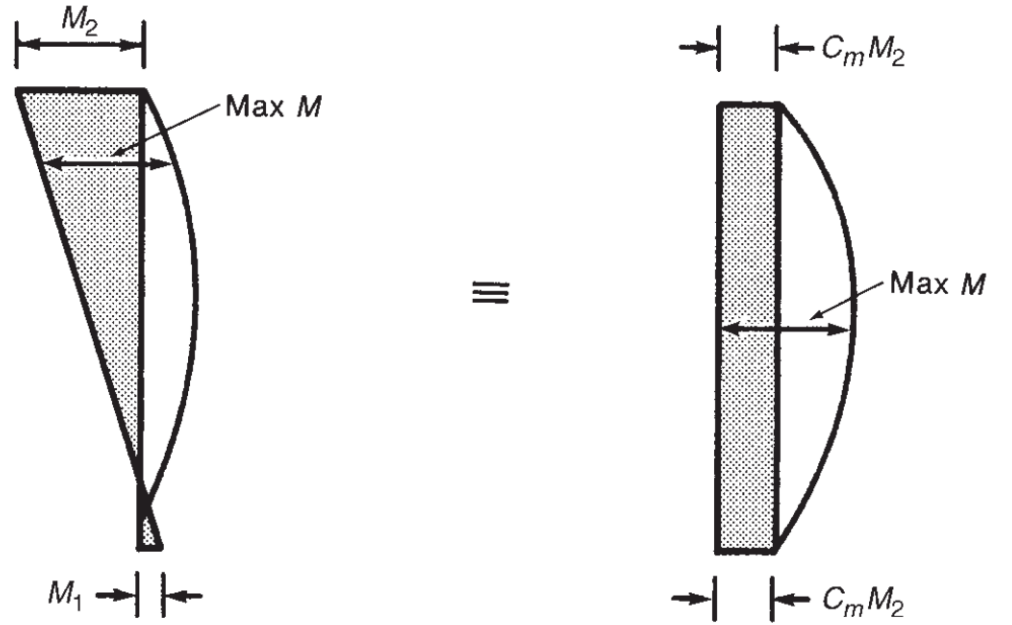
δ = Nonsway-moment magnifier

Effect of Unequal End Moments on the Slender Column Strength

In the moment-magnifier design procedure, the column subjected to unequal end moments shown in Fig. 12-13a is replaced with a similar column subjected to equal moments of $C_m M_2$ at both ends, as shown in Fig. 12-13b. The moments $C_m M_2$ are chosen so that the maximum magnified moment is the same in both columns. The expression for the *equivalent moment factor* C_m was originally derived for use in the design of steel beam-columns [12-4] and was adopted without change for concrete design [12-1] (ACI Code Section 10.10.6.4):

$$C_m = 0.6 + 0.4 \frac{M_1}{M_2} \qquad (12\text{-}14)$$

Eq. 10-16)



(a) Actual moments at failure.

(b) Equivalent moments at failure.

Effect of Unequal End Moments: The C_m Factor

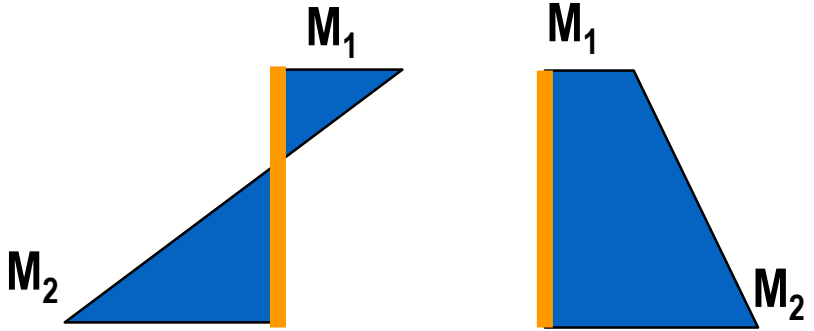
- The Moment and Stress Amplification Factors are derived on the basis of pin-ended columns with single moment curvature.

  (C_m = 1.0)

- For other Moment Distribution, the correction factor C_m needs to be computed to modify the stress amplification.

  C_m = 0.4 to 1.0

$$C_m = 0.6 + 0.4\frac{M1}{M2} \geq 0.4$$



M_1 is the smaller End Moment

M_2 is the larger End Moment

Effect of Unequal End Moments on the Slender Column Strength

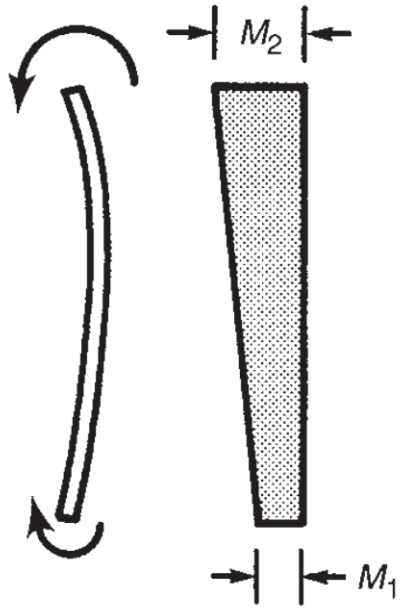
$$C_m = 0.6 + 0.4\frac{M_1}{M_2} \qquad (12\text{-}14)$$

(ACI Eq. 10-16)

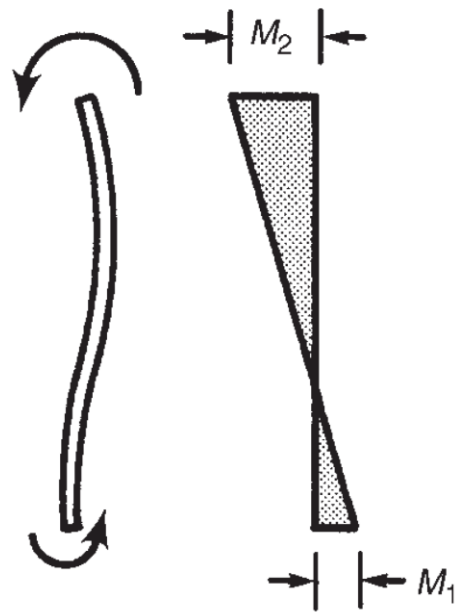
Equation (12-14) was originally derived using the ratio of end eccentricities (e_1/e_2) rather than the ratio of end moments (M_1/M_2). When it was changed to end moments, the original eccentricity sign convention was retained. Thus, in Eq. (12-14), M_1 and M_2 are the smaller and larger end moments, respectively, calculated from a conventional first-order elastic analysis. The sign convention for the ratio M_1/M_2 is illustrated in Fig. 12-13c and d. If the moments M_1 and M_2 cause single curvature bending without a point of contraflexure between the ends, as shown in Fig. 12-13c, M_1/M_2 is positive. If the moments M_1 and M_2 bend the column in double curvature with a point of zero moment between the two ends, as shown in Fig. 12-13d, that M_1/M_2 is negative.

Equation (12-14) applies only to hinged columns or columns in braced frames, loaded with axial loads and end moments. In all other cases, including columns subjected to transverse loads between their ends and concentrically loaded columns (no end moment), C_m is taken equal to 1.0 (ACI Code Section 10.10.6.4). The term C_m is not included in the equation for the moment magnifier for unbraced (sway) frames.

Effect of Unequal End Moments on the Slender Column Strength

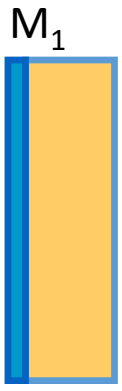


(c) Single curvature column.
$0 \leq M_1/M_2 \leq 1.0$



(d) Double curvature column.
$-1 \leq M_1/M_2 \leq 0$

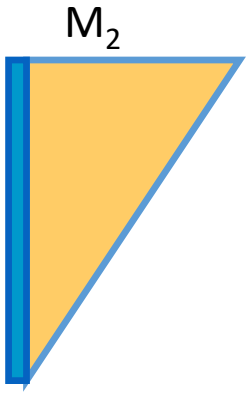
More about C_m Factor



$M_1 = M$
$M_2 = M$

$$\frac{M_1}{M_2} = 1$$

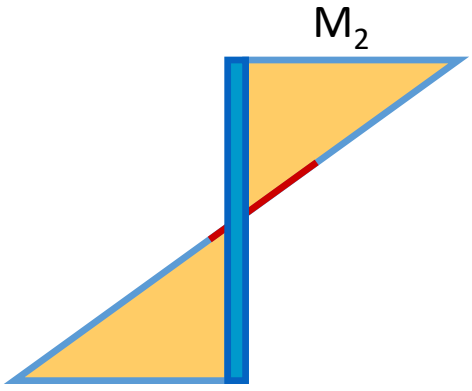
C_m = 1.0



$M_1 = 0$
$M_2 = M$

$$\frac{M_1}{M_2} = 0$$

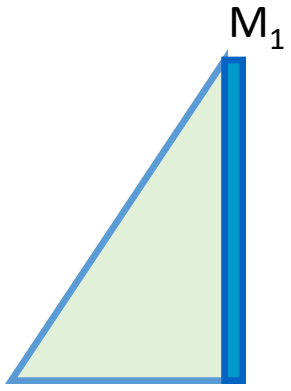
C_m = 0.6



$M_1 = -M$
$M_2 = M$

$$\frac{M_1}{M_2} = -1$$

C_m = 0.4



$M_1 = 0$
$M_2 = -M$

$$\frac{M_1}{M_2} = 0$$

C_m = 0.6

Limiting Slenderness Ratios for Slender Columns

Most columns in structures are sufficiently short and stocky to be unaffected by slenderness effects. To avoid checking slenderness effects for all columns, ACI Code Section 10.10.1 allows slenderness effects to be ignored in the case of columns in sway frames if,

$$\frac{k\ell_u}{r} \leq 22 \qquad \text{(12-20a)}$$

(ACI Eq. 10-6)

and in nonsway frames if,

$$\frac{k\ell_u}{r} \leq 34 - 12\frac{M_1}{M_2} \leq 40 \qquad \text{(12-20b)}$$

(ACI Eq. 10-7)

In Eq. (12-20), k refers to the effective length factor, which is 1.0 for a pin-ended column, ℓ_u is the unsupported height (ACI Code Section 10.10.1.1.), and r is the radius of gyration, taken as $0.3h$ for rectangular sections and $0.25h$ for circular sections (ACI Code Section 10.10.1.2). For other shapes, the value of r can be calculated from the area and moment of inertia of the cross section. By definition, $r = \sqrt{I/A}$. The sign convention for M_1/M_2 is given in Fig. 12-13.

Pin-ended columns having slenderness ratios less than the right-hand side of Eq. (12-20) should have a slender-column strength that is 95 percent or more of the short-column strength.

Definition of Nonsway and Sway Frames

The preceding discussions were based on the assumption that frames could be separated into nonsway (braced) frames and sway (unbraced) frames. A column may be considered to be nonsway in a given direction if the lateral stability of the structure as a whole is provided by walls, bracing, or buttresses designed to resist all lateral forces in that direction. A column may be considered to be part of a sway frame in a given plane if all resistance to lateral loads comes from bending of the columns.

In actuality, there is no such thing as a "completely braced" frame, and no clear-cut boundary exists between nonsway and sway frames. Some frames are clearly unbraced, as, for example, the frames shown in Fig. 12-25b and c. Other frames are connected to shear walls, elevator shafts, and so on, which clearly restrict the lateral movements of the frame, as shown in Fig. 12-25a. Because no wall is completely rigid, however, there will always be some lateral movement of a braced frame, and hence some $P\Delta$ moments will result from the lateral deflections.

For the purposes of design, a story or a frame can be considered "nonsway" if horizontal displacements do not significantly reduce the vertical load capacity of the structure. This criterion could be restated as follows: a frame can be considered "nonsway" if the $P\Delta$ moments due to lateral deflections are small compared with the first-order moments due to lateral loads. ACI Code Section 10.10.5.1 allows designers to assume that a frame is nonsway if the increase in column-end moments due to second-order effects does not exceed 5 percent of the first-order moments.

Definition of Nonsway and Sway Frames

Alternatively, ACI Code Section 10.10.5.2 allows designers to assume that a story in a frame is nonsway if

$$Q = \frac{\Sigma P_u \Delta_o}{V_{us} \ell_c} \leq 0.05 \qquad \text{(12-21)}$$

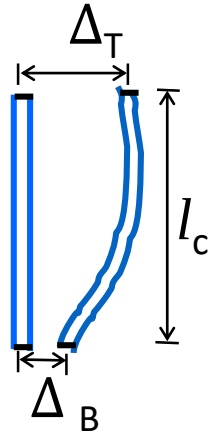
(ACI Eq. 10-10)

where Q is the *stability index*, ΣP_u is the total vertical load in all the columns and walls in the story in question, V_{us} is the shear in the story due to factored lateral loads, Δ_o is the first-order relative deflection between the top and bottom of that story *due to* V_{us}, and ℓ_c is the height of the story measured from center to center of the joints above and below the story.

Sway Frames

- Appreciable relative movement of two ends of column.

$$Sway\Delta_0 = \frac{|\Delta_T - \Delta_B|}{l_c}$$



- Sway Limits

$$a)\ EI_{Bracing\ walls} > 6EI_{Columns}$$

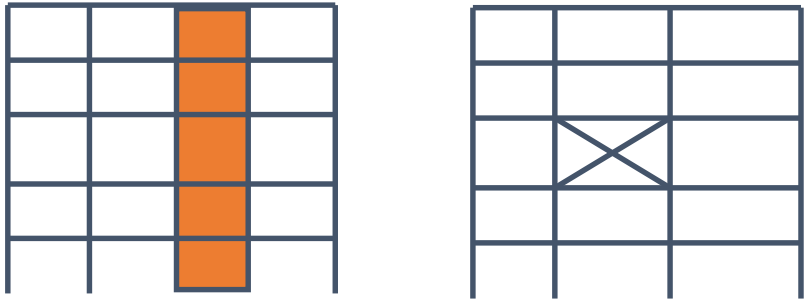
$$b)\ \frac{P_U \Delta_0}{V_U l_C} < 0.05$$

$$c)\ \frac{M_m}{M} < 1.05$$

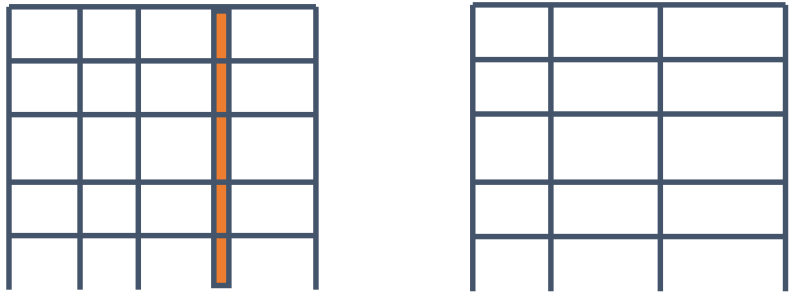
⇨ Frame considered as “Non-Sway”

More on Sway Frames

- Braced Column (Non-Sway)



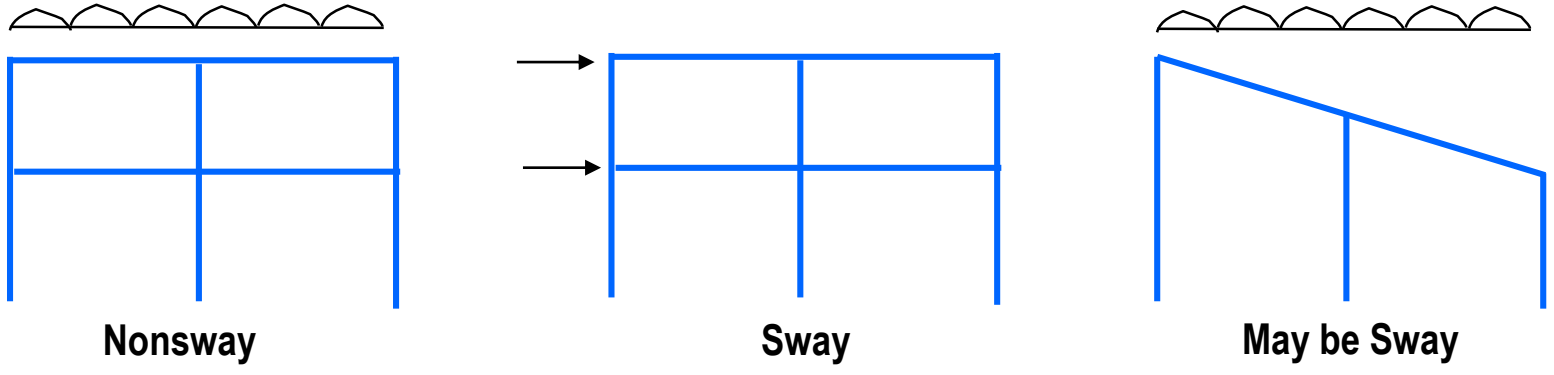
- Unbraced Column (Sway)



- Most building columns may be considered “Non-Sway” for gravity loads.
- More than 40% of columns in buildings are “Non-Sway” for lateral loads.
- Moment Magnification for “Sway” case is more significant, more complicated and more important.

More on Sway Frames

Sway is dependent upon the structural configuration as well as type of loading.



ACI 318 Moment Magnification Method

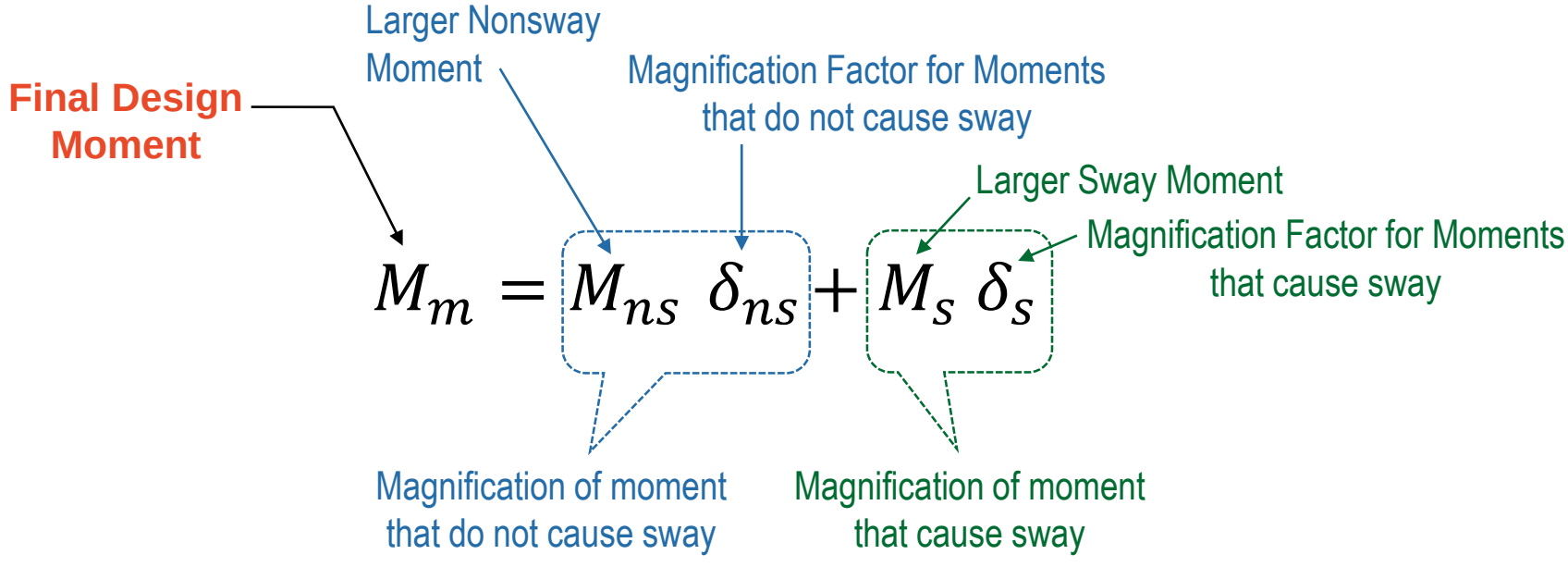
The Moment Magnifier Method – ACI 318

- The Moment Magnifier Method
  - An Approximate Method to account for Slenderness Effects.
  - May be used instead of P-Δ Analysis.
  - Not to be used when Kl/r > 100.
  - Separate Magnification for Sway and Non-Sway Load Cases.
  - Separate Magnification Factors for moment about each axis.
  - Moment magnification generally 1.2 to 2.5 times.
  - Mostly suitable for building columns.

When to use Moment Magnification?

- According to ACI 318-11 Code:
  - For Braced Frames (Non-sway)
    - Kl/r > 34-12(M1b/M2b)
  - For Un-braced Frames (Sway)
    - Kl/r > 22
  - Or When Secondary Moments become Significant
- These provisions do not consider other factors, such as P, lateral deflection, lateral loads, section material or properties.

Moment Magnification – ACI 318



Determination of Design Moment: ACI 318

Calculation of δ_{ns} (Non-Sway)

Moment curvature Coefficient

$$\delta_{ns} = \frac{C_m}{1 - \frac{P_u}{0.75 P_C}}$$

Applied column load

Critical buckling load

Equation 10-12
ACI 318-11

Flexural Stiffness

$$P_C = \frac{\pi^2 (EI)}{(Kl_U)^2}$$

Effective Length Factor

Equation 10-13
ACI 318-11

Determination of δ_s (Sway)

Equation 10-20 ACI 318-11

Sway Quotient

$$a)\ \delta_s = \frac{1}{1-Q} \geq 1.0 \quad \text{where } Q = \frac{\sum P_u \Delta_0}{V_u l_c}$$

Equation 10-10 ACI 318-11

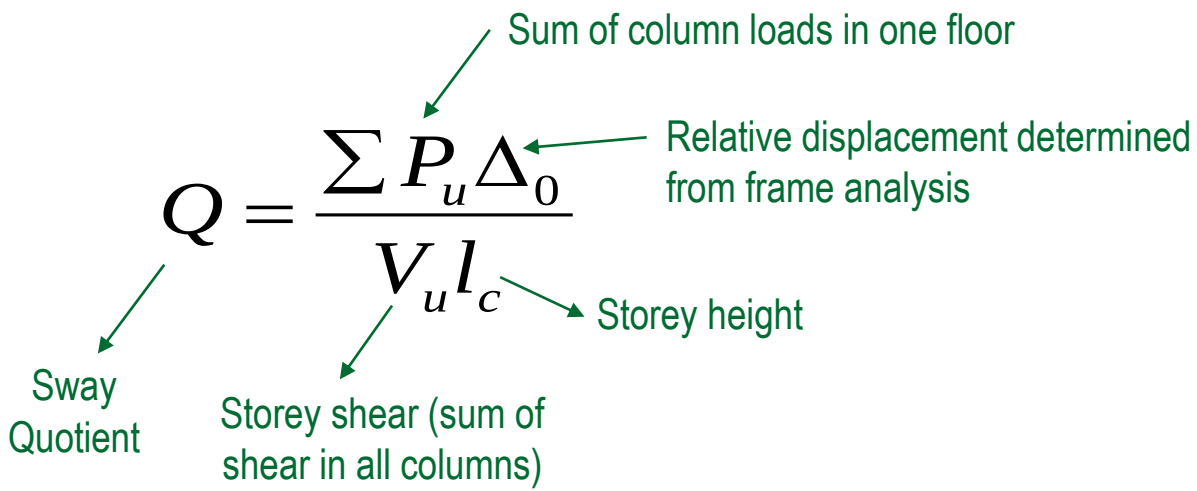
If $\delta_s > 1.5$ *then*

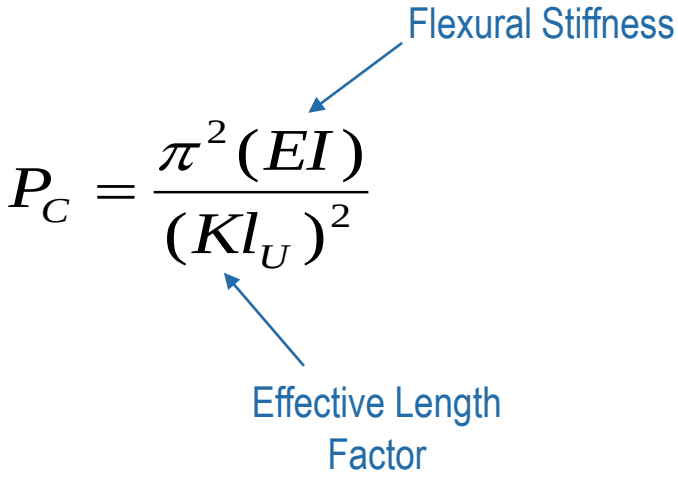
$$b)\ \delta_s = \frac{1}{1-\dfrac{\sum P_u}{0.75 \sum P_c}} \geq 1$$

Equation 10-21 ACI 318-11

Sum of Critical Buckling Load of all columns in floor

Determination of δ_s (Sway)





Equation 10-13
ACI 318-11

Moment Magnification Factors – ACI 318

For Non-sway Frames (Very rigid or braced)	$\delta_s = 1.0$ $\delta_{ns} \geq 1.0$
For Sway Frames (Open frames, not braced, depends on loads also)	$\delta_s \geq 1.0$ $\delta_{ns} \geq 1.0$

Moment Magnification Summary

Larger Non- Sway Moment

Larger Sway Moment

Final Design Moment $M_m = M_{ns}\ \delta_{ns} + M_s\ \delta_s$

Equation 10-12 ACI 318-11

$$\delta_{ns} = \frac{C_m}{1 - \frac{P_u}{0.75\,P_C}}$$

$$a)\ \ \delta_s = \frac{1}{1 - \frac{\sum P_u \Delta_0}{V_u l_c}} \geq 1.0$$

$$If\ \ \delta_s > 1.5\ \ then$$

$$b)\ \ \delta_s = \frac{1}{1 - \frac{\sum P_u}{0.75 \sum P_c}} \geq 1$$

Equation 10-16 ACI 318-11

$$C_m = 0.6 + 0.4\frac{M1}{M2} \geq 0.4 \qquad P_C = \frac{\pi^2 (EI)}{(Kl_U)^2}$$

Flexural Stiffness for determining the Euler Buckling Load

Design Values of EI for the Computation of the Critical Loads of Individual Columns

ACI Code Section 10.10.6.1 gives the following equations for the computation of *EI* in calculating the critical load of an individual column:

$$EI = \frac{0.2E_cI_g + E_sI_{se}}{1 + \beta_{dns}} \qquad (12\text{-}15)$$

(ACI Eq. 10-14)

$$EI = \frac{0.40E_cI_g}{1 + \beta_{dns}} \qquad (12\text{-}16)$$

(ACI Eq. 10-15)

In both equations,

E_c, E_s = moduli of elasticity of the concrete (ACI Code Section 8.5.1) and the steel, respectively

I_g = gross moment of inertia of the concrete section about its centroidal axis, ignoring the reinforcement

I_{se} = moment of inertia of the reinforcement about the centroidal axis of the concrete section

The term $(1 + \beta_{dns})$ reflects the effect of creep on the column deflections and is discussed later.

ACI Moment Magnifier Design Procedure for Columns in Non-sway Frames

If a column is in a nonsway frame, then design following the moment magnifier method involves ACI Code Sections 10.10.5 and 10.10.6. The specific equations and requirements for determining the magnified moments are:

1. Length of column. The unsupported length, ℓ_u, is defined in ACI Code Section 10.10.1.1 as the clear distance between members capable of giving lateral support to the column. For a pin-ended column it is the distance between the hinges.

2. Effective length. ACI Commentary Section R10.10.1 states that the effective length factor, k, can be taken conservatively as 1.0 for columns in nonsway frames. A procedure for estimating k-values for columns in nonsway and sway frames will be given in Section 12-4.

3. Radius of gyration. For a rectangular section $r = 0.3h$, and for a circular section, $r = 0.25h$. (See ACI Code Section 10.10.1.2.) For other sections, r can be calculated from the area and moment of inertia of the gross concrete section as $r = \sqrt{I_g/A_g}$.

4. Consideration of slenderness effects. For columns in nonsway frames, ACI Code Section 10.10.1 allows slenderness to be neglected if $k\ell_u/r$ satisfies Eq. (12-20b). The sign convention for M_1/M_2 is given in Fig. 12-13.

5. Minimum moment. ACI Code Section 10.10.6.5 requires that the maximum end moment on the column, M_2, not be taken less than

$$M_{2,\min} = P_u(0.6 + 0.03h) \qquad (12\text{-}22)$$

(ACI Eq. 10-17)

where the 0.6 and h are in inches. When M_2 is less than $M_{2,\min}$, C_m shall either be taken equal to 1.0 or evaluated from the actual end moments.

ACI Moment Magnifier Design Procedure for Columns in Non-sway Frames

6. Moment-magnifier equation. ACI Code Section 10.10.6 states that the columns shall be designed for the factored axial load, P_u, and the magnified moment, M_c, defined by

$$M_c = \delta_{ns} M_2 \tag{12-23}$$

(ACI Eq. 10-11)

The subscript *ns* refers to nonsway. The moment M_2 is defined as the larger end moment acting on the column. ACI Code Section 10.10.6 goes on to define δ_{ns} as

$$\delta_{ns} = \frac{C_m}{1 - P_u/(0.75P_c)} \geq 1.0 \tag{12-24}$$

(ACI Eq. 10-12)

where

$$C_m = 0.6 + 0.4\frac{M_1}{M_2} \tag{12-14}$$

(ACI Eq. 10-16)

$$P_c = \frac{\pi^2 EI}{(k\ell_u)^2} \tag{12-25}$$

(ACI Eq. 10-13)

and

$$EI = \frac{0.2E_c I_g + E_s I_{se}}{1 + \beta_{dns}} \tag{12-15}$$

(ACI Eq. 10-14)

$$EI = \frac{0.40E_c I_g}{1 + \beta_{dns}} \tag{12-16}$$

(ACI Eq. 10-15)

$$\beta_{dns} = \frac{\text{maximum factored sustained (dead) axial load in column}}{\text{total factored axial load in column}}$$

ACI Moment Magnifier Design Procedure for Columns in Non-sway Frames

If P_u exceeds $0.75P_c$ in Eq. (12-24), δ_{ns} will be negative, and the column would be unstable. Hence, if P_u exceeds $0.75P_c$, the column cross section must be enlarged. Further, if δ_{ns} exceeds 2.0, strong consideration should be given to enlarging the column cross section, because calculations for such columns are very sensitive to the assumptions being made.

$$\delta_{ns} = \frac{C_m}{1 - P_u/(0.75P_c)} \geq 1.0 \qquad (12\text{-}24)$$

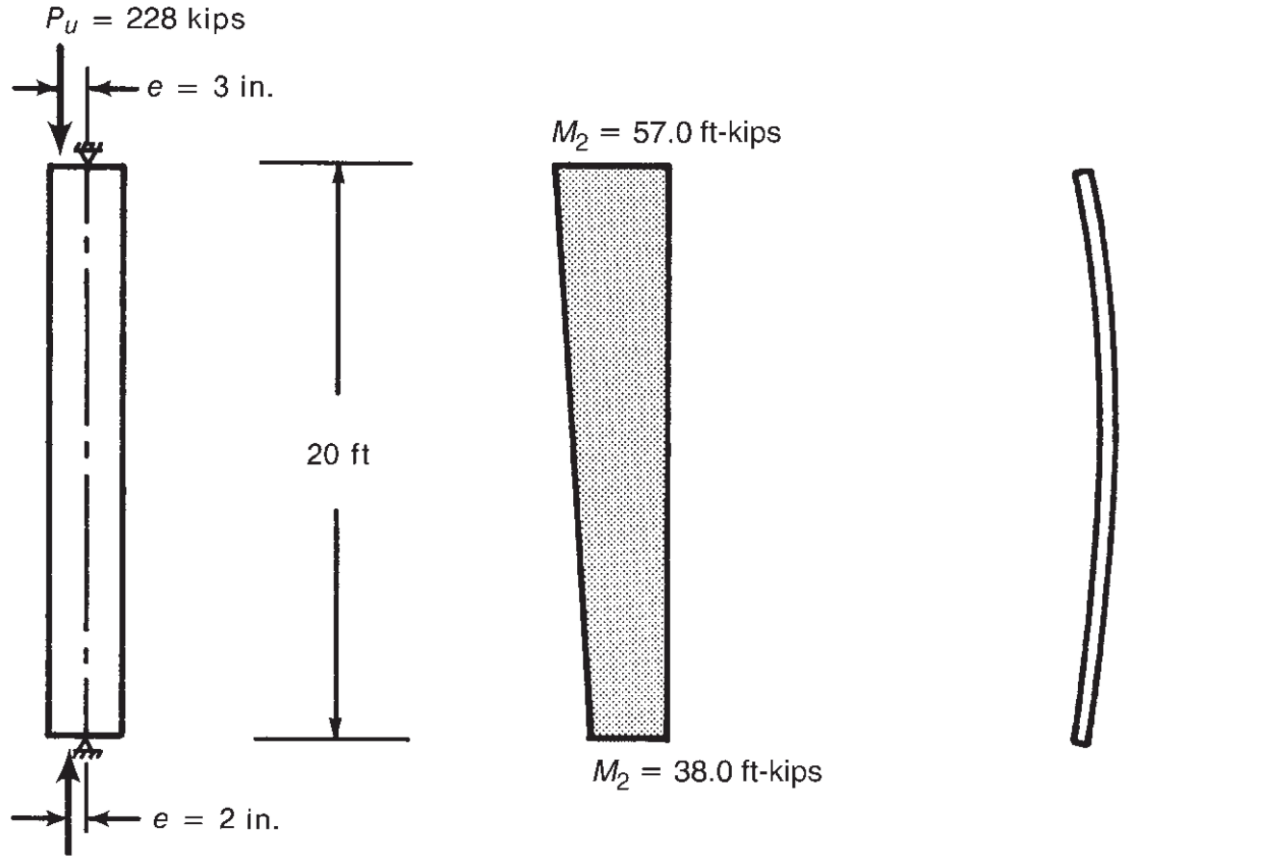
(ACI Eq. 10-12)

Design of Pin-ended Columns in Non-sway Frames

Solved Example

Design of a Slender Pin-Ended Column (Nonsway)

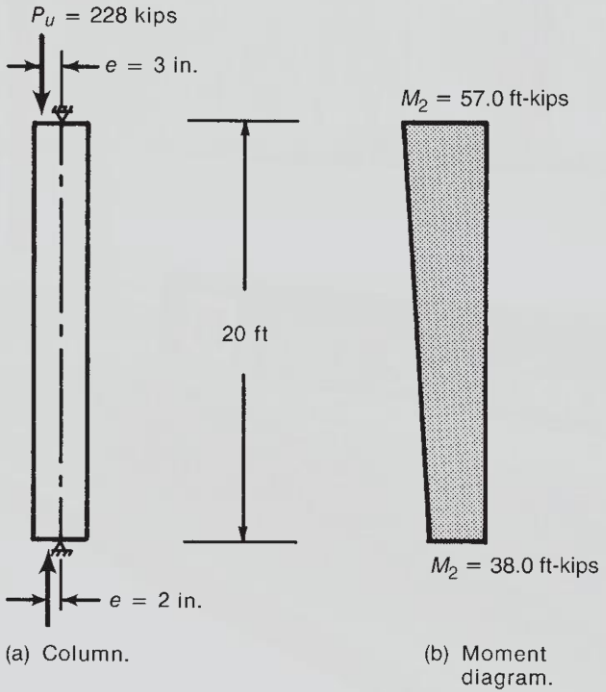
Design a 20-ft-tall column to support an unfactored dead load of 90 kips and an unfactored live load of 75 kips. The loads act at an eccentricity of 3 in. at the top and 2 in. at the bottom, as shown in Fig. 12-16. Use $f'_c = 4000$ psi and $f_y = 60{,}000$ psi. Use the load combinations and strength-reduction factors from ACI Code Sections 9.2 and 9.3.



(a) Column. (b) Moment diagram. (c) Deflected shape.

Solved Example

Design of a Slender Pin-Ended Column (Nonsway)



1. Compute the factored loads and moments and M_1/M_2.

$$P_u = 1.2D + 1.6L$$
$$= 1.2 \times 90 \text{ kips} + 1.6 \times 75 \text{ kips} = 228 \text{ kips}$$

The moment at the top is

$$M = P_u \times e$$
$$= 228 \text{ kips} \times \frac{3 \text{ in.}}{12 \text{ in.}} = 57.0 \text{ kip-ft}$$

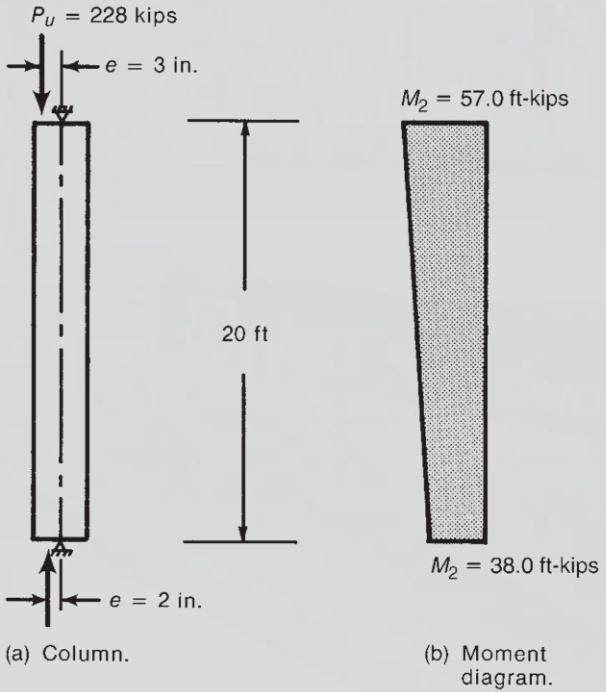
The moment at the bottom is

$$M = 228 \text{ kips} \times \frac{2 \text{ in.}}{12 \text{ in.}}$$
$$= 38.0 \text{ kip-ft}$$

By definition, M_2 is the larger end moment in the column. Therefore, $M_2 = 57.0$ kip-ft and $M_1 = 38.0$ kip-ft. The ratio M_1/M_2 is taken to be positive, because the column is bent in single curvature (see Fig. 12-16c). Thus $M_1/M_2 = 0.667$.

Solved Example

Design of a Slender Pin-Ended Column (Nonsway)



2. Estimate the column size. From (11-19a), assuming that $\rho_g = 0.015$,

$$A_{g(\text{trial})} \geq \frac{P_u}{0.40(f'_c + f_y\rho_g)}$$

$$A_{g(\text{trial})} \geq \frac{228 \times 1000}{0.40(4000 + 60{,}000 \times 0.015)}$$

$$= 116 \text{ in.}^2$$

This suggests that a 12 in. × 12 in. column would be satisfactory. It should be noted that Eq. (11-19a) was derived for short columns and will underestimate the required sizes of slender columns.

3. Is the column slender? From Eq. (12-20b), a column in a nonsway frame is short if

$$\frac{k\ell_u}{r} < 34 - 12\frac{M_1}{M_2} \leq 40 \qquad (12\text{-}20b)$$

For the 12 in. × 12 in. section selected in step 2, $k = 1.0$ because the column is pin-ended, and where $r = 0.3h = 0.3 \times 12$ in. $= 3.6$ in.,

$$\frac{k\ell_u}{r} = \frac{1.0 \times 240 \text{ in.}}{3.6 \text{ in.}} = 66.7$$

For $M_1/M_2 = 0.667$,

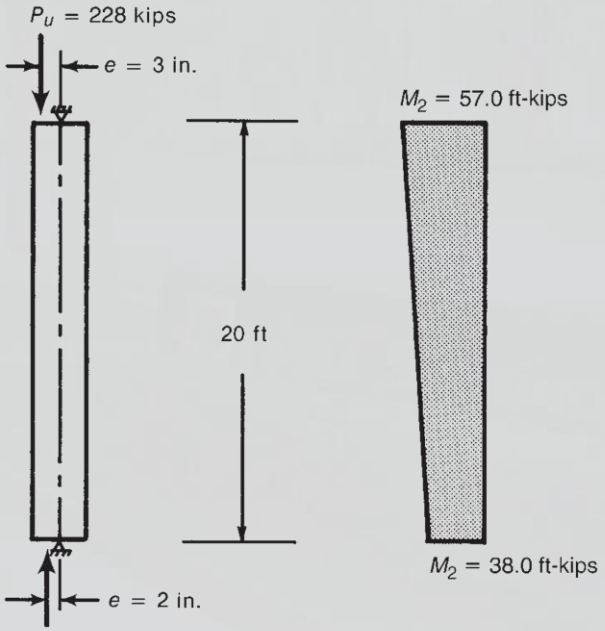
$$34 - 12\frac{M_1}{M_2} = 34 - 12 \times 0.667 = 26$$

Because $k\ell_u/r = 66.7$ exceeds 26, the column is quite slender. This suggests that the 12 in. × 12 in. section probably is inadequate.

Using increments of 2 in., we shall select a 16-in.-by-16 in.-section for the first trial.

Solved Example

Design of a Slender Pin-Ended Column (Nonsway)



4. Check whether the moments are less than the minimum. ACI Code Section 10.10.6.5 requires that a braced column be designed for a minimum eccentricity of $0.6 + 0.03h = 1.08$ in. Because the maximum end eccentricity exceeds this, design for the moments from step 1.

5. Compute *EI*. At this stage, the area of reinforcement is not known. Additional calculations are needed before it is possible to use Eq. (12-15) to compute *EI*, but Eq. (12-16) can be used.

$$EI = \frac{0.40E_cI_g}{1 + \beta_{dns}} \tag{12-16}$$

where

$$E_c = 57{,}000\sqrt{f'_c} = 3.60 \times 10^6 \text{ psi}$$

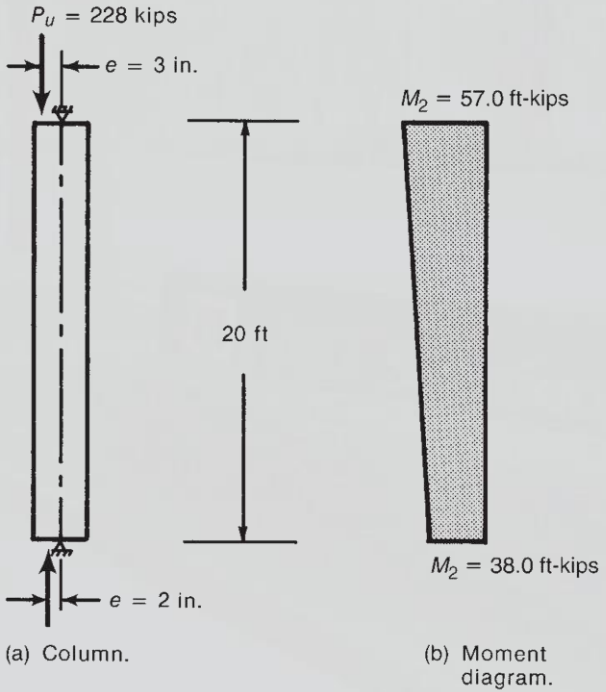
$$I_g = bh^3/12 = 5460 \text{ in.}^4$$

The term β_{dns} is the ratio of the factored sustained (dead) load to the total factored axial load:

$$\beta_{dns} = \frac{1.2 \times 90}{228} = 0.474 \tag{12-26a}$$

Solved Example

Design of a Slender Pin-Ended Column (Nonsway)



Thus,

$$EI = 0.4 \times \frac{3.60 \times 10^6 \text{ psi} \times 5460 \text{ in.}^4}{(1 + 0.474)}$$
$$= 5.33 \times 10^9 \text{ lb-in.}^2$$

Generally, one would use Eq. (12-15) or Eq. (12-19) if ρ_g exceeded about 0.02.

6. Compute the magnified moment. From Eq. (12-23),

$$M_c = \delta_{ns} M_2$$

where

$$\delta_{ns} = \frac{C_m}{1 - P_u/0.75P_c} \geq 1.0 \qquad (12\text{-}24)$$

$$C_m = 0.6 + 0.4\frac{M_1}{M_2} \qquad (12\text{-}14)$$
$$= 0.6 + 0.4 \times 0.667 = 0.867$$

$$P_c = \frac{\pi^2 EI}{(k\ell_u)^2} \qquad (12\text{-}25)$$

where $k = 1.0$ because the column is pin-ended.

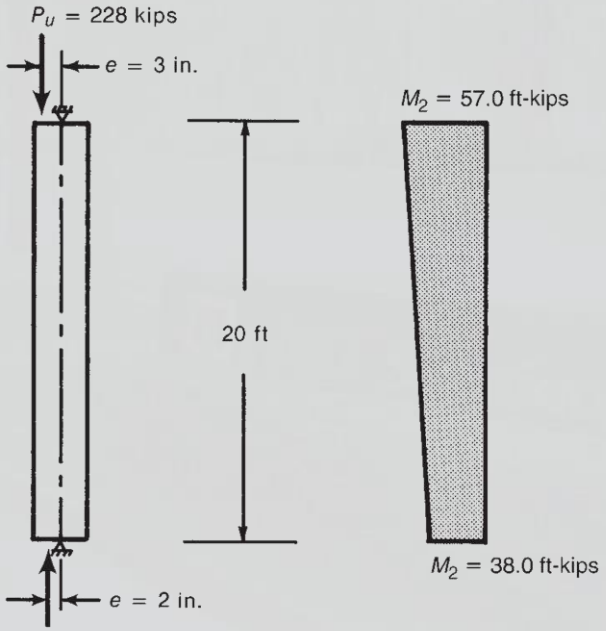
$$P_c = \frac{\pi^2 \times 5.33 \times 10^9 \text{ lb-in.}^2}{(1.0 \times 240 \text{ in.})^2} = 913{,}000 \text{ lb}$$
$$= 913 \text{ kips}$$

and

$$\delta_{ns} = \frac{0.867}{1 - 228/(0.75 \times 913)} \geq 1.0$$
$$= 1.30$$

Solved Example

Design of a Slender Pin-Ended Column (Nonsway)



Normally, if δ_{ns} exceeds 1.75 to 2.0, a larger cross section should be selected. Continuing without selecting a larger column, the magnified moment is

$$M_c = 1.30 \times 57.0 \text{ kip-ft} = 74.1 \text{ kip-ft}$$

Because δ_{ns} is small, **use a 16-in.-by-16-in. square section.**

7. Select the column reinforcement. We will use the tied-column interaction diagrams in Appendix A assuming an equal distribution of longitudinal bars in two opposite faces of the column. The parameters required for entering the interaction diagrams are

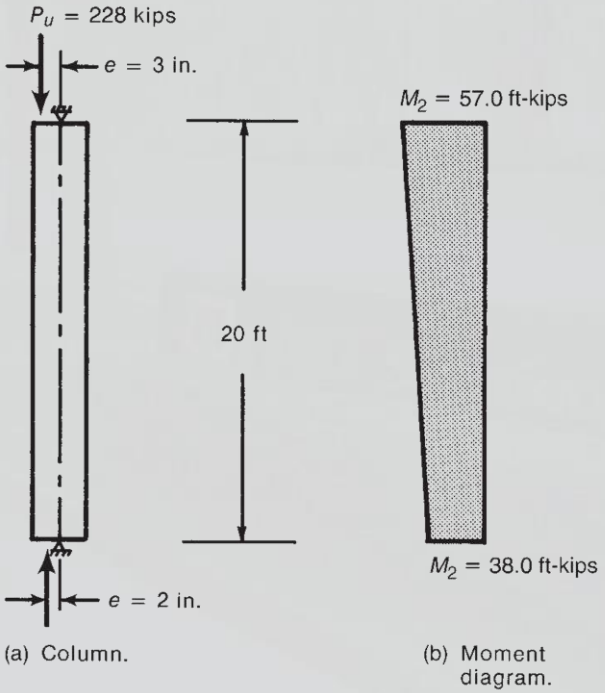
$$\gamma \cong \frac{16 \text{ in.} - 2 \times 2.5 \text{ in.}}{16 \text{ in.}} = 0.69$$

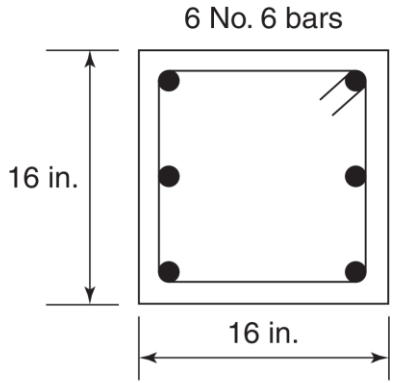
Assuming $\phi P_n = P_u = 228$ kips,

$$\frac{\phi P_n}{bh} = \frac{228 \text{ kips}}{16 \text{ in.} \times 16 \text{ in.}} = 0.89 \text{ ksi}$$

Solved Example

Design of a Slender Pin-Ended Column (Nonsway)





Assuming $\phi M_n = M_c = 74.1$ kip-ft $= 889$ kip-in.,

$$\frac{\phi M_n}{bh^2} = \frac{889 \text{ kip-in.}}{(16 \text{ in.})^3} = 0.22 \text{ ksi}$$

From both Fig. A-6a ($\gamma = 0.60$) and Fig. A-6b ($\gamma = 0.75$), the required value for ρ_g is less than 0.01. Therefore, to satisfy the minimum column longitudinal-reinforcement ratio from ACI Code Section 10.9.1, use $\rho_g = 0.01$. Thus,

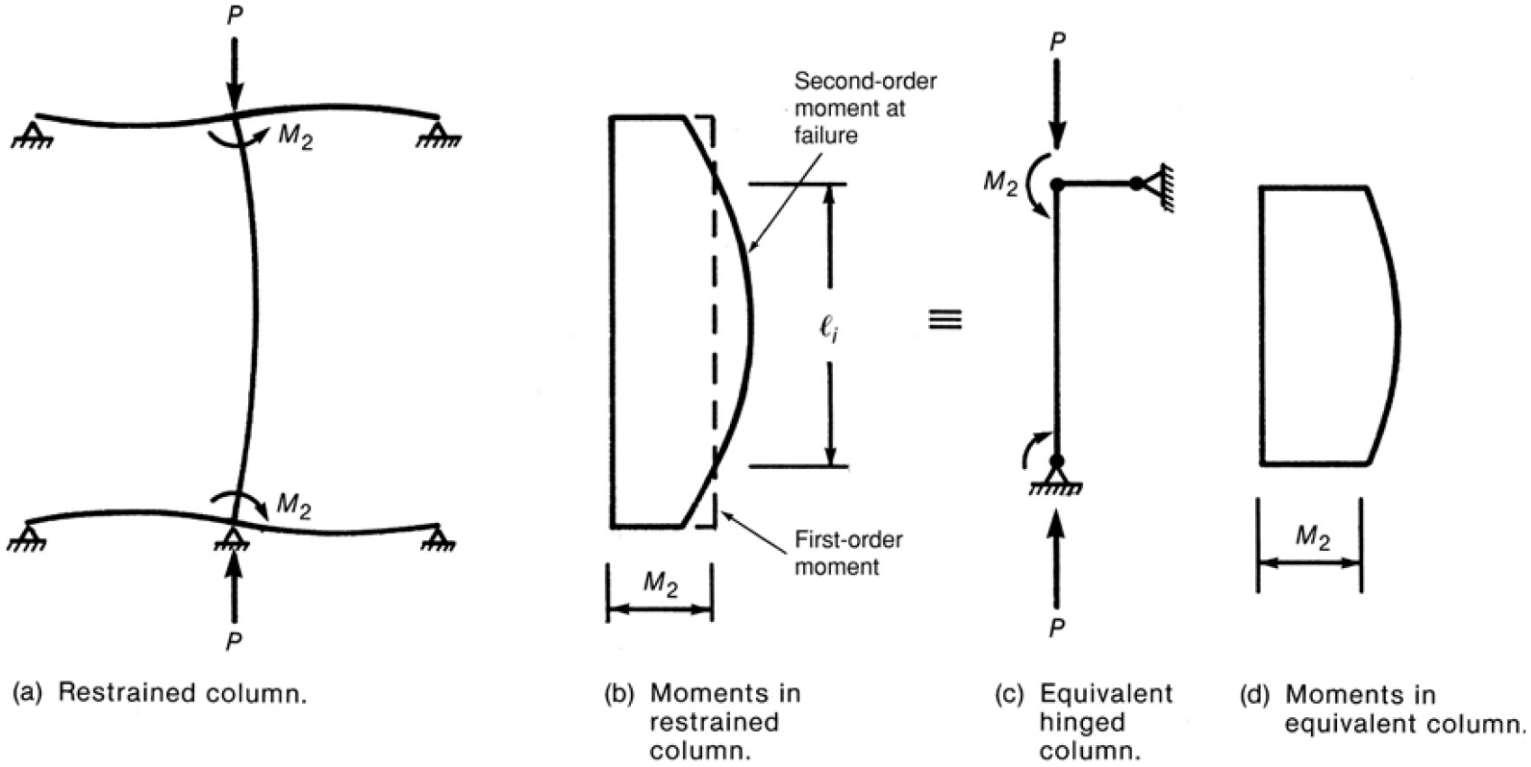
$$A_{st}(\text{req'd}) = 0.01 \times A_g = 0.01 \times (16 \text{ in.})^2 = 2.56 \text{ in.}^2$$

Use six No. 6 bars, $A_{st} = 6 \times 0.44 \text{ in.}^2 = 2.64 \text{ in.}^2$ In summary, **use a 16 in. × 16 in. column section with six No. 6 bars**, as shown in Fig. 12-17. This section design would be very conservative if we were designing a short column, but the slenderness of the column has required the use of this larger section. ■

Behavior and Design of Restrained Columns in Non-sway Frames

Design Approximation for the Effect of End Restraints in Nonsway Frames

Replacement of restrained column with an equivalent hinged column for design.



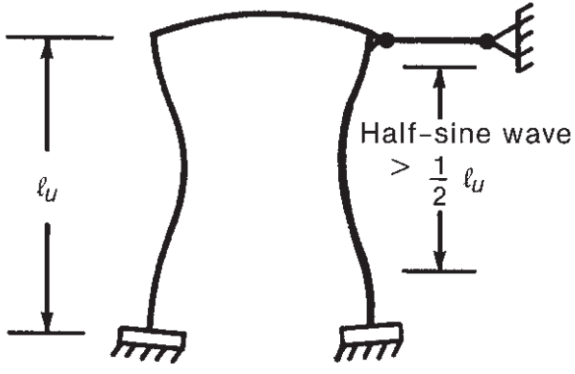
(a) Restrained column.

(b) Moments in restrained column.

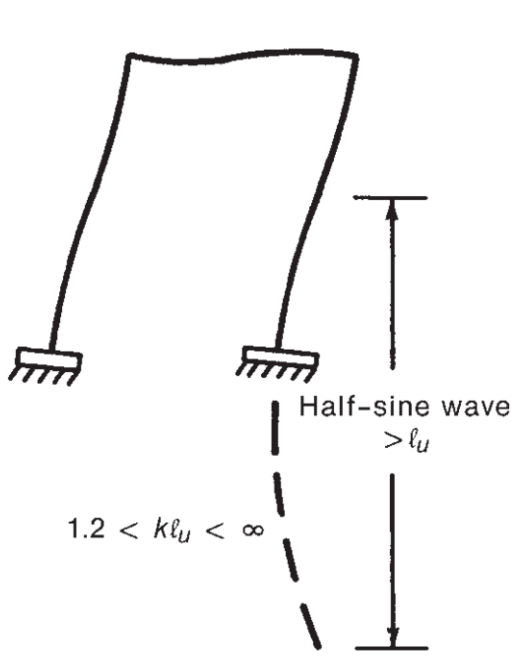
(c) Equivalent hinged column.

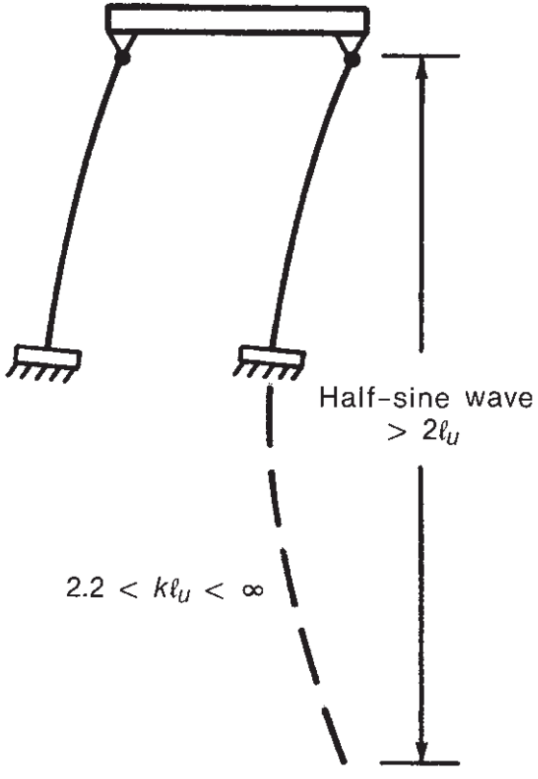
(d) Moments in equivalent column.

Effective lengths of columns in frames with foundation rotations



(a) Nonsway frame.





(b) and (c) Sway frames.

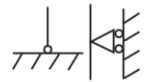
The actual value of k for an elastic column is a function of the relative stiffnesses of the beams and columns at each end of the column.

Effective Length of Columns in Nonsway Frames

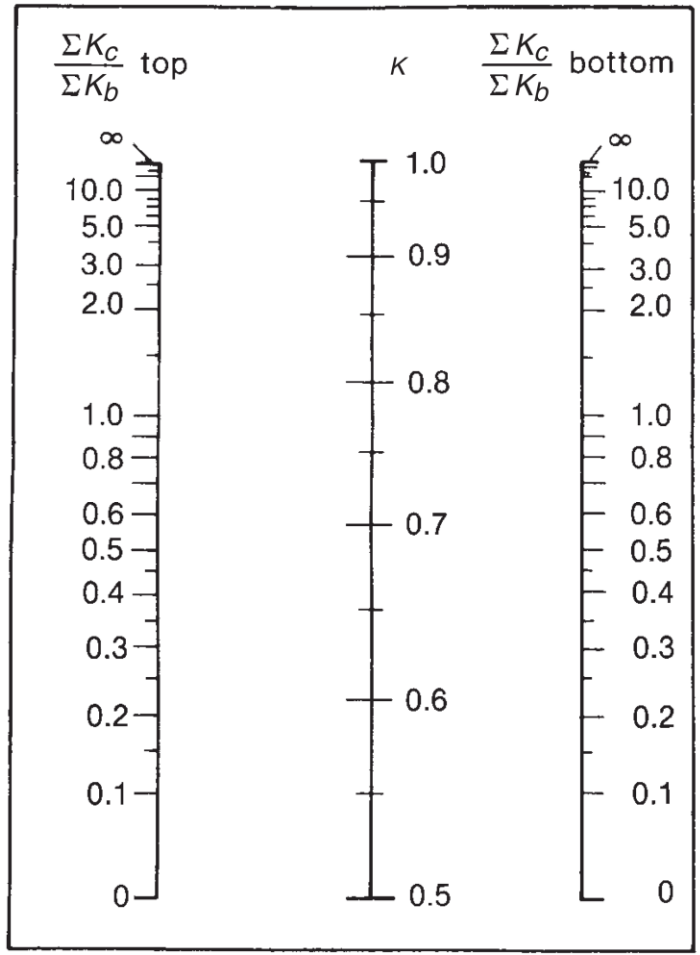
- The effective length of a column is defined as the length of an equivalent pin-ended column having the same buckling load.
- When a pin-ended column buckles, its deflected shape is a half-sine wave.
- The effective length of a restrained column is taken equal to the length of a complete half-sine wave in the deflected shape.

$$\psi = \frac{\Sigma\,(E_c I_c/\ell_c)}{\Sigma\,(E_b I_b/\ell_b)}$$

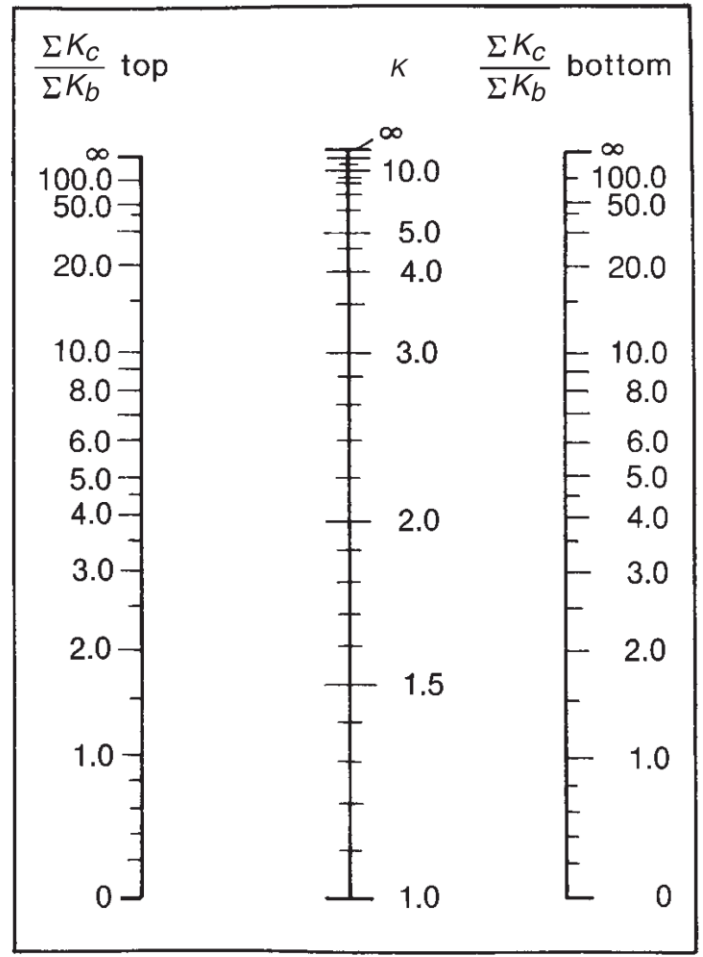
TABLE 12-2 Effective-Length Factors for Nonsway (Braced) Frames

Top	*k*				
Hinged 	0.70	0.81	0.91	0.95	1.00
Elastic $\psi = 3.1$ 	0.67	0.77	0.86	0.90	0.95
Elastic, flexible $\psi = 1.6$ 	0.65	0.74	0.83	0.86	0.91
Stiff $\psi = 0.4$ 	0.58	0.67	0.74	0.77	0.81
Fixed 	0.50	0.58	0.65	0.67	0.70
	 Fixed	 Stiff	 Elastic, flexible	 Elastic	 Hinged
Bottom					

Nomographs for effective length factors



(a) Nonsway frames.

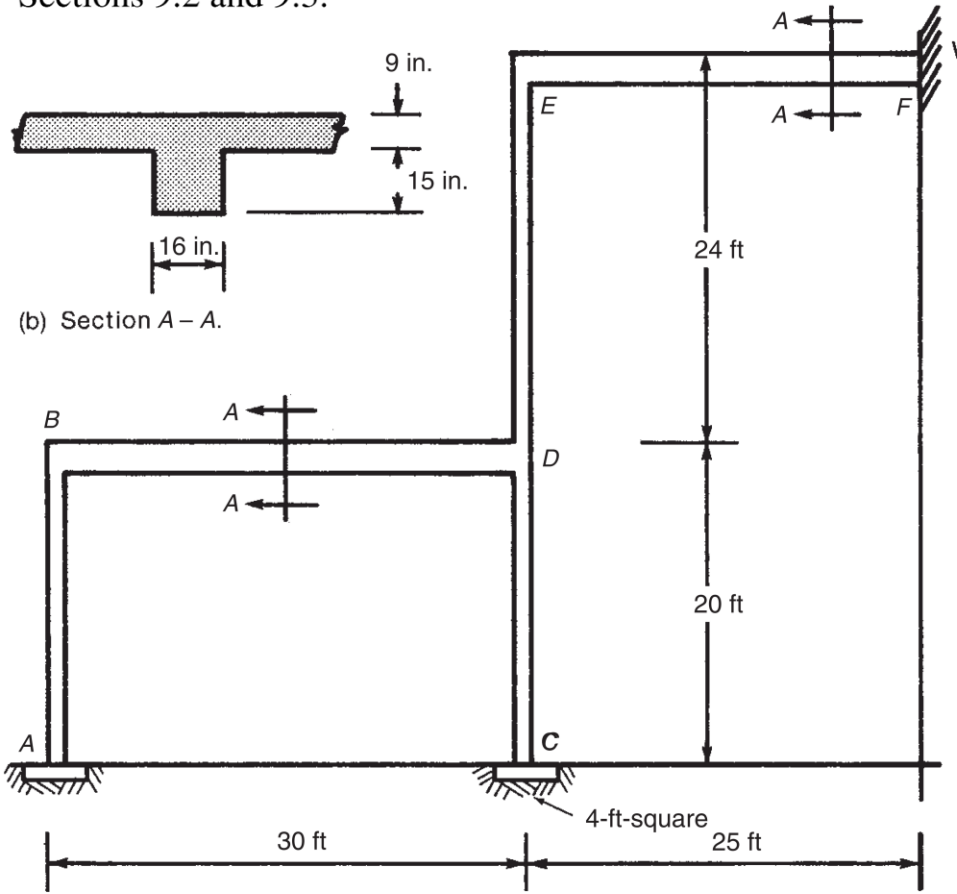


(b) Sway frames.

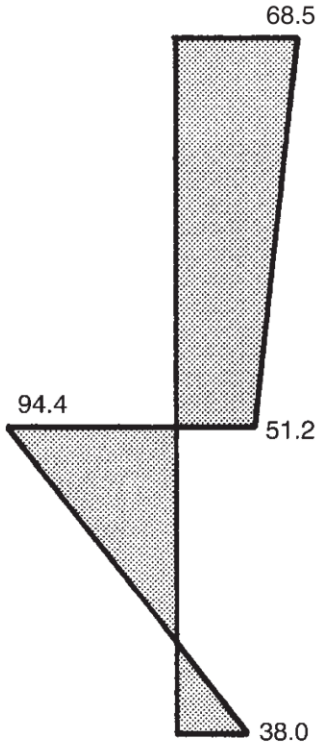
Solved Example

Design of a Slender Restrained Column in Nonsway (Braced) Frame

Figure 12-28 shows part of a typical frame in an industrial building. The frames are spaced 20 ft apart. The columns rest on 4-ft-square footings. The soil bearing capacity is 4000 psf. Design columns *C–D* and *D–E*. Use $f'_c = 4000$ psi and $f_y = 60{,}000$ psi for beams and columns. Use load combinations and strength-reduction factors from ACI Code Sections 9.2 and 9.3.



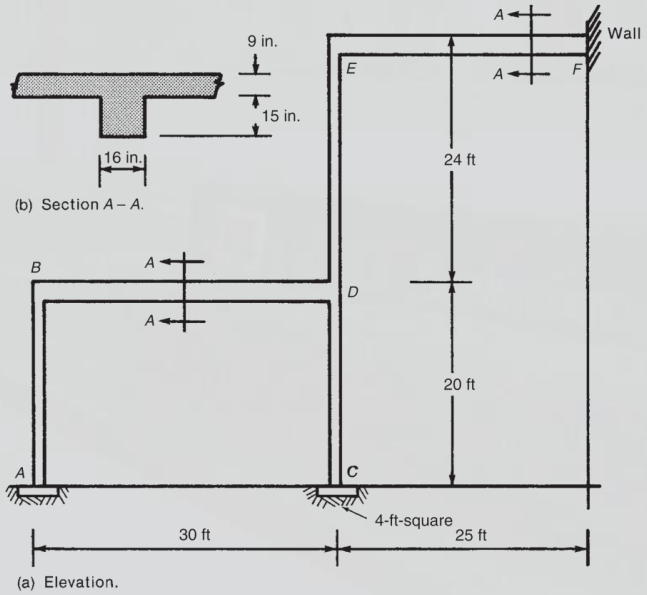
(a) Elevation.



(c) Moments in columns *CD* and *DE*, (kip-ft).

Solved Example

Design of a Slender Restrained Column in Nonsway (Braced) Frame



1. Calculate the column loads from a frame analysis.

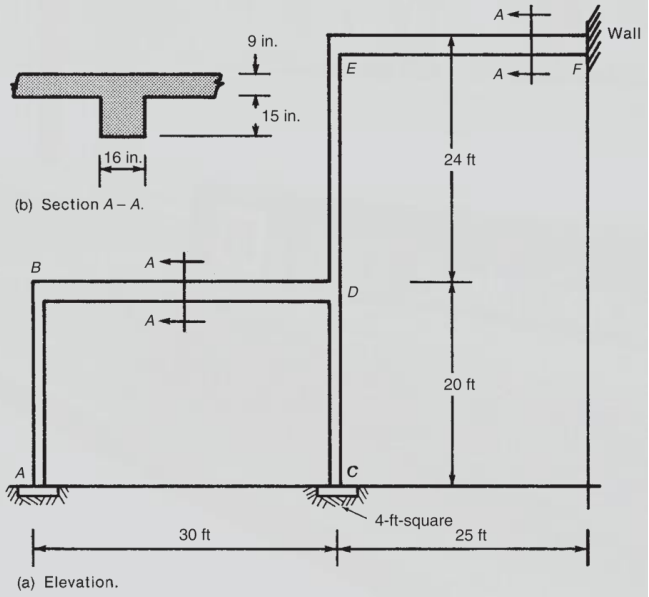
A first-order elastic analysis of the frame shown in Fig. 12-28 (loads acting on frame members are not shown) gave the forces and moments in the table.

	Column *CD*	Column *DE*
Service loads, P	Dead = 80 kips	Dead = 50 kips
	Live = 24 kips	Live = 14 kips
Service moments at	Dead = −60 kip-ft	Dead = 42.4 kip-ft
tops of columns	Live = −14 kip-ft	Live = 11.0 kip-ft
Service moments at	Dead = −21 kip-ft	Dead = −32.0 kip-ft
bottoms of columns	Live = −8 kip-ft	Live = −8 kip-ft

Clockwise moments on the ends of members are positive. All wind forces are assumed to be resisted by the end walls of the building.

Solved Example

Design of a Slender Restrained Column in Nonsway (Braced) Frame



2. Determine the factored loads.

(a) Column *CD*:

$$P_u = 1.2 \times 80 + 1.6 \times 24 = 134 \text{ kips}$$

$$\text{Moment at top} = 1.2 \times -60 + 1.6 \times -14 = -94.4 \text{ kip-ft}$$

$$\text{Moment at bottom} = 1.2 \times -21 + 1.6 \times -8 = -38.0 \text{ kip-ft}$$

The factored-moment diagram is shown in Fig. 12-28c. By definition (ACI Code Section 2.1), M_2 is always positive, and M_1 is positive if the column is bent in single curvature (Fig. 12-13c and d). Because column *CD* is bent in double curvature (Fig. 12-28c), M_1 is negative. Thus, for slender column design, $M_2 = +94.4$ kip-ft and $M_1 = -38.0$ kip-ft.

(b) Column *DE*:

$$P_u = 82.4 \text{ kips}$$

$$\text{Moment at top} = +68.5 \text{ kip-ft}$$

$$\text{Moment at bottom} = -51.2 \text{ kip-ft}$$

Thus $M_2 = +68.5$ kip-ft and $M_1 = +51.2$ kip-ft. M_1 is positive because the column is in single curvature.

3. Make a preliminary selection of the column size. From Eq. (11-19a) for $\rho_g = 0.015$,

$$A_{g(\text{trial})} \geq \frac{P_u}{0.40(f'_c + f_y\rho_g)}$$

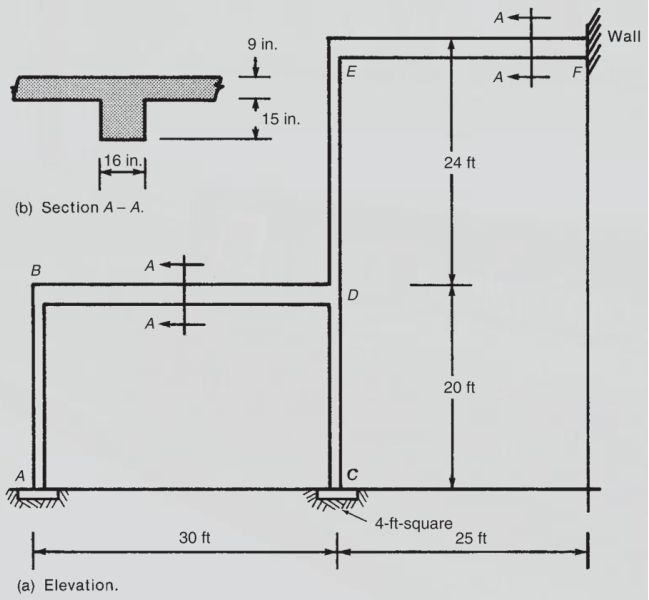
$$= \frac{134}{0.40(4 + 0.015 \times 60)}$$

$$= 68.4 \text{ in.}^2 \text{ or } 8.3 \text{ in. square}$$

Because of the anticipated slenderness effects and because of the large moments, we will take a larger column. Try 14 in. × 14 in. columns throughout.

Solved Example

Design of a Slender Restrained Column in Nonsway (Braced) Frame



4. Are the columns slender? From ACI Code Section 10.10.1, a column in a braced frame is short if $k\ell_u/r$ is less than $34 - 12M_1/M_2$, which has an upper limit of 40.

(a) Column *CD*:

$$\ell_u = 20 \text{ ft} - 2 \text{ ft} = 18 \text{ ft}$$
$$= 216 \text{ in.}$$
$$r = 0.3 \times 14 \text{ in.} = 4.2 \text{ in.}$$

From Table 12-2, $k \cong 0.77$. Thus,

$$\frac{kl_u}{r} \cong \frac{0.77 \times 216}{4.2} = 39.6$$

From Eq. (12-20b),

$$34 - 12\left(\frac{M_1}{M_2}\right) = 34 - 12\left(-\frac{38.0}{94.4}\right) = 38.8$$

Because $39.6 > 38.8$, column *CD* is just slender.

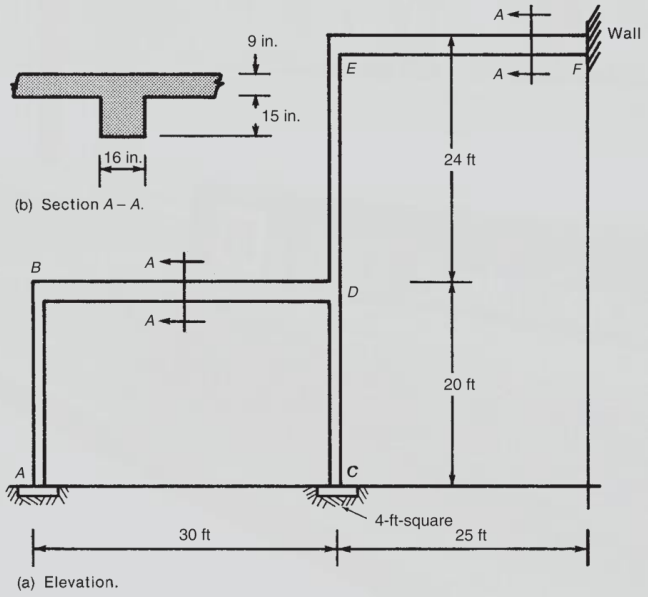
(b) Column *DE*:

$$\ell_u = 24 \text{ ft} - 2 \text{ ft} = 264 \text{ in.}$$
$$k \cong 0.86$$
$$\frac{kl_u}{r} \cong \frac{0.86 \times 264}{4.2} = 54.1 > 40$$

Thus, column DE is also slender.

Solved Example

Design of a Slender Restrained Column in Nonsway (Braced) Frame



5. Check whether the moments are less than the minimum. ACI Code Section 10.10.6.5 requires that braced slender columns be designed for a minimum eccentricity of $(0.6 + 0.03h)$ in. For 14-in. columns, this is 1.02 in. Thus, column *CD* must be designed for a moment M_2 of at least

$$P_u e_{\min} = 134 \text{ kips} \times 1.02 \text{ in.} = 11.4 \text{ kip-ft}$$

and column *DE* for a moment of at least 7.0 kip-ft. Because the actual moments exceed these values, the columns shall be designed for the actual moments.

6. Compute *EI*. Because the reinforcement is not known at this stage of the design, we can use Eq. (12-16) to compute *EI*. From Eq. (12-16),

$$EI = \frac{0.40\, E_c I_g}{1 + \beta_{dns}}$$

where

$$E_c = 57{,}000\sqrt{f'_c} = 3.60 \times 10^6 \text{ psi}$$

$$I_g = 14^4/12 = 3200 \text{ in.}^4$$

$$0.40\, E_c I_g = 4.61 \times 10^9 \text{ lb-in}^2.$$

(a) Column *CD*:

$$\beta_{dns} = \frac{1.2 \times 80}{134} = 0.716 \qquad (12\text{-}26a)$$

$$EI = \frac{4.61 \times 10^9}{1 + 0.716} = 2.69 \times 10^9 \text{ lb-in.}^2$$

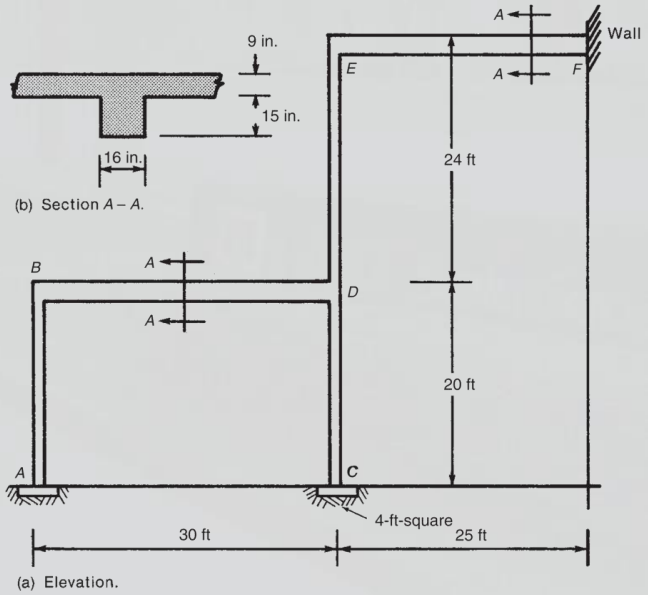
(b) Column *DE*:

$$\beta_{dns} = \frac{1.2 \times 50}{82.4} = 0.728$$

$$EI = \frac{4.61 \times 10^9}{1.728} = 2.67 \times 10^9 \text{ lb-in}^2.$$

Solved Example

Design of a Slender Restrained Column in Nonsway (Braced) Frame



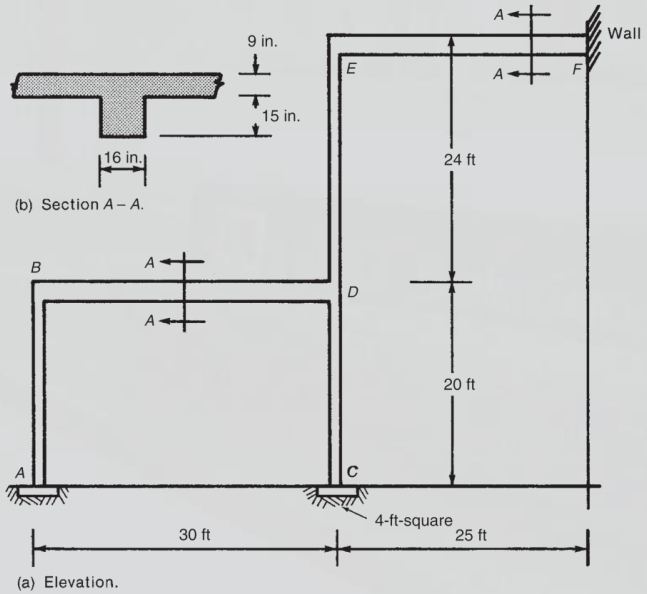
7. Compute the effective-length factors. Two methods of estimating the effective-length factors, k, have been presented. In this example, we calculate k by both methods, to illustrate their use. In practice, only one of these procedures would be used in a given set of calcultions. We begin with

$$\psi = \frac{\Sigma E_c I_c/\ell_c}{\Sigma E_b I_b/\ell_b} \qquad \text{(12-29a)}$$

where values for I_c and I_b can be taken as those given in ACI Code Section 10.10.4.1. Thus, $I_c = 0.70\, I_g(\text{col})$, and $I_b = 0.35\, I_g(\text{beam})$. For the assumed column section, use $I_c = 0.70 \times 14^4/12 = 2240$ in.^{4}. For the beam section shown in Fig. 12-28b, the governing condition in ACI Code Section 8.12.2 gives the effective flange width as being equal to one-fourth of the beam span, i.e., 90 in. Using that width gives $I_g = 36{,}600$ in.^{4}, so $I_b = 0.35 \times 36{,}600 = 12{,}800$ in.^{4}. As noted earlier in this text, I_g for the web portion of the beam section is equal to approximately one-half of I_g for the full-beam section (web and flange), and it often is used as an estimate of the cracking moment of inertia, I_{cr}, for the beam section. Thus, for the analysis used here, $0.35 I_g(\text{beam})$ is equal to approximately $0.70\, I_g(\text{web}) = 0.7 \times 16 \times 24^3/12 = 12{,}900$ in.^{4}. This approximation could be used at this stage to save the time required to determine the effective flange width and then calculate the moment of inertia of a flanged section.

Solved Example

Design of a Slender Restrained Column in Nonsway (Braced) Frame



In Eq. (12-29a), ℓ_c and ℓ_b are the spans of the column and beam, respectively, measured center-to-center from the joints in the frame.

(a) Column *DE*: The value of ψ at E is

$$\psi_E = \frac{E_c \times 2240/(24 \times 12)}{E_b \times 12{,}800/300} = 0.182\,\frac{E_c}{E_b}$$

where $E_c = E_b$. Thus, $\psi_E = 0.182$. The value of ψ at D is (for $E_c = E_b$):

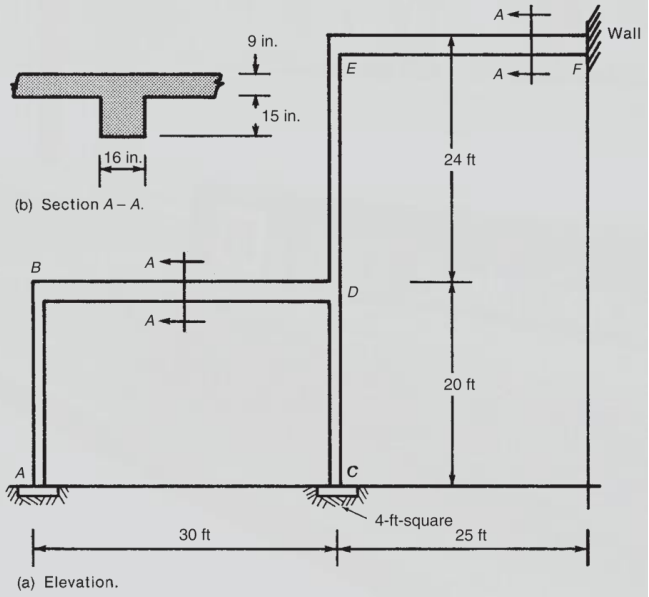
$$\psi_D = \frac{E_c \times 2240/288 + E_c \times 2240/240}{E_b \times 12{,}800/360} = 0.481$$

The value of k from Fig. 12-26a is 0.63. The value of k from Table 12-2 was approximately 0.86.

As was pointed out in the discussion of Fig. 12-26, the effective-length nomographs tend to underestimate the values of k for beam columns in practical frames [12-18]. Because Table 12-2 gives reasonable values without the need to calculate ψ, it has been used to compute k in this example. Thus, we shall use $k = 0.86$ for column *DE*.

Solved Example

Design of a Slender Restrained Column in Nonsway (Braced) Frame



(b) Column *CD*: The value of ψ at D is

$$\psi_D = 0.481$$

The column is restrained at C by the rotational resistance of the soil under the footing and is continuous at D. From Eq. (12-34),

$$\psi = \frac{4E_cI_c/\ell_c}{I_fk_s}$$

where I_f is the moment of inertia of the contact area between the footing and the soil and k_s is the coefficient of subgrade reaction, which is 160 psi/in. from Fig. 12-27. Using , $I_c = 1.0\,I_g$ (not $0.7I_g$) and assuming a footing depth of 2 ft, so ℓ_c in Eq. (12-34) is 22 ft = 264 in., results in

$$I_f = \frac{48^4}{12} = 442{,}000 \text{ in.}^4$$

$$\psi_c = \frac{4 \times 3.60 \times 10^6 \text{ lb/in.}^2 \times 3200 \text{ in.}^4/264 \text{ in.}}{442{,}000 \text{ in.}^4 \times 160 \text{ lb/in.}^3}$$

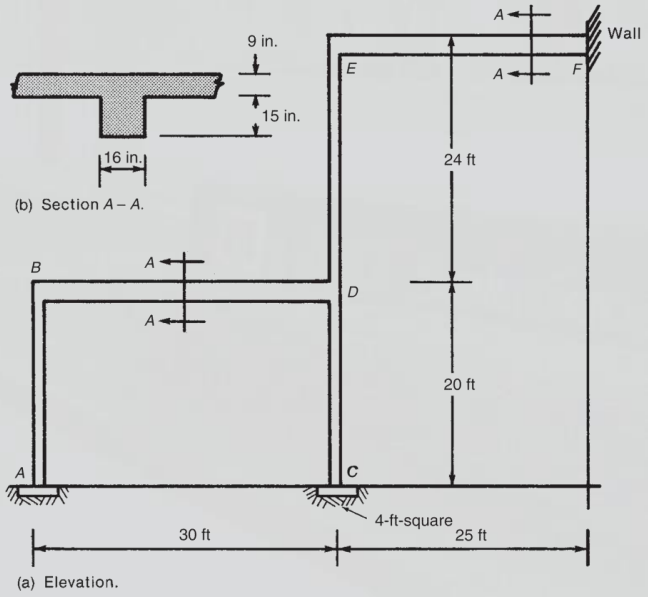
$$= 2.47$$

From Fig. 12-26: $k = 0.77$
From Table 12-2: $k \cong 0.77$

Use $k = 0.77$ for column *CD*.

Solved Example

Design of a Slender Restrained Column in Nonsway (Braced) Frame



8. Compute the magnified moments. From Eq. (12-23),

$$M_c = \delta_{ns} M_2 \tag{12-23}$$

where

$$\delta_{ns} = \frac{C_m}{1 - (P_u/0.75P_c)} \geq 1.0 \tag{12-24}$$

(a) Column *CD*:

$$C_m = 0.6 + 0.4\left(-\frac{38.0}{94.4}\right) \tag{12-14}$$

$$= 0.438$$

$$P_c = \frac{\pi^2 EI}{(k\ell_u)^2} \tag{12-25}$$

$$\ell_u = 18 \text{ ft} = 216 \text{ in}$$

Using EI from step 6 and $k = 0.77$,

$$P_c = \frac{\pi^2 \times 2.69 \times 10^9 \text{ lb-in.}^2}{(0.77 \times 216 \text{ in.})^2}$$

$$= 960{,}000 = 960 \text{ kips}$$

$$\delta_{ns} = \frac{0.438}{1 - 134/(0.75 \times 960)}$$

$$= 0.538 \geq 1.0$$

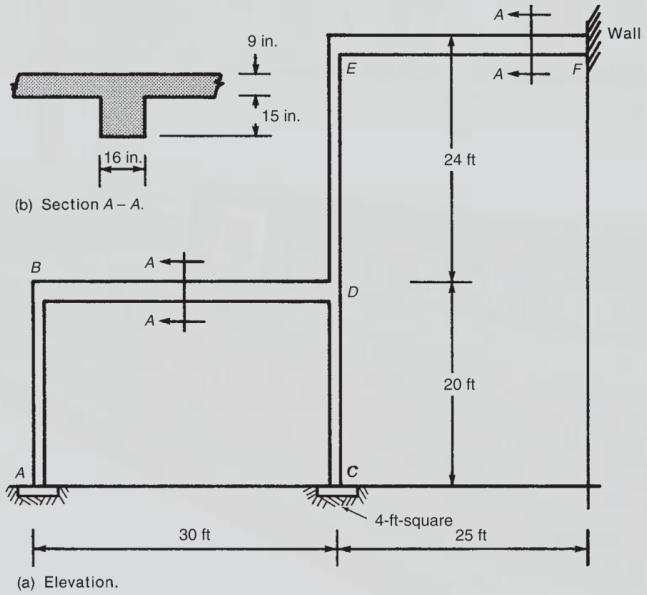
Therefore, $\delta_{ns} = 1.0$. This means that the section of maximum moment remains at the end of the column, so that

$$M_c = 1.0 \times 94.4 = 94.4 \text{ kip-ft}$$

Column *CD* is designed for $P_u = 134$ kips and $M_u = M_c = 94.4$ kip-ft.

Solved Example

Design of a Slender Restrained Column in Nonsway (Braced) Frame



(b) Column *DE*:

$$C_m = 0.6 + 0.4\left(\frac{51.2}{68.5}\right) = 0.900$$

$$P_c = \frac{\pi^2 \times 2.67 \times 10^9}{(0.86 \times 264)^2} = 511 \text{ kips}$$

$$\delta_{ns} = \frac{0.900}{1 - \dfrac{82.4}{0.75 \times 511}} \geq 1.0$$

$$= 1.15$$

This column is affected by slenderness, so

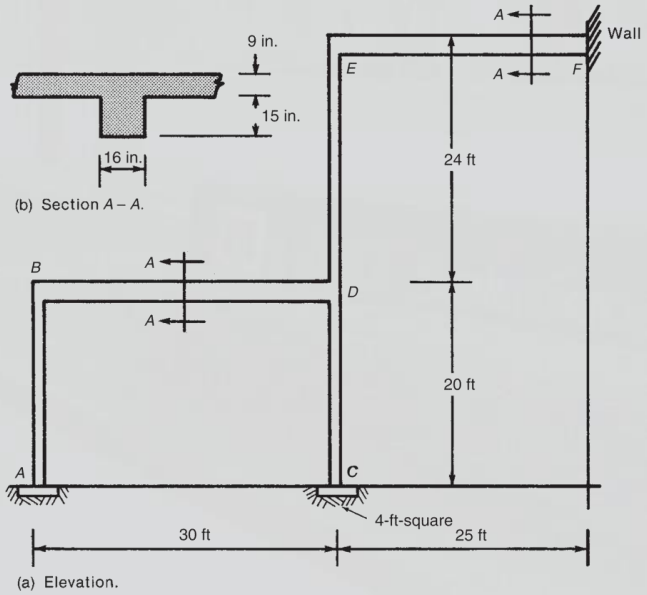
$$M_c = 1.15 \times 68.5 = 78.8 \text{ kip-ft}$$

Column *DE* is designed for $P_u = 82.4$ kips and $M_u = M_c = 78.8$ kip-ft.

For both columns *CD* and *DE* the magnified moments are less than 1.4 times the first-order moments, as required by ACI Code Section 10.10.2.1.

Solved Example

Design of a Slender Restrained Column in Nonsway (Braced) Frame



9. Select the column reinforcement. Column *CD* is carrying both a higher axial load and a higher moment, and thus, it will govern the design of the column section. We will use the tied-column interaction diagrams in Appendix A, assuming an equal distribution of longitudinal bars in two opposite faces of the column. The parameters required for entering the interaction diagrams are

$$\gamma \cong \frac{14 \text{ in.} - 2 \times 2.5 \text{ in.}}{14 \text{ in.}} = 0.64 \cong 0.60$$

Assuming $\phi P_n = P_u = 134$ kips,

$$\frac{\phi P_n}{bh} = \frac{134 \text{ kips}}{14 \text{ in.} \times 14 \text{ in.}} = 0.68 \text{ ksi}$$

Assuming $\phi M_n = M_c = 94.4$ kip-ft $= 1130$ kip-in.,

$$\frac{\phi M_n}{bh^2} = \frac{1130 \text{ kip-in.}}{(14 \text{ in.})^3} = 0.41 \text{ ksi}$$

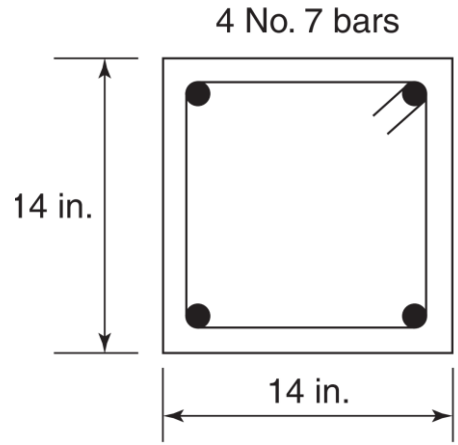
From Fig. A-6a ($\gamma = 0.60$), the required value for ρ_g is approximately 0.012. Thus,

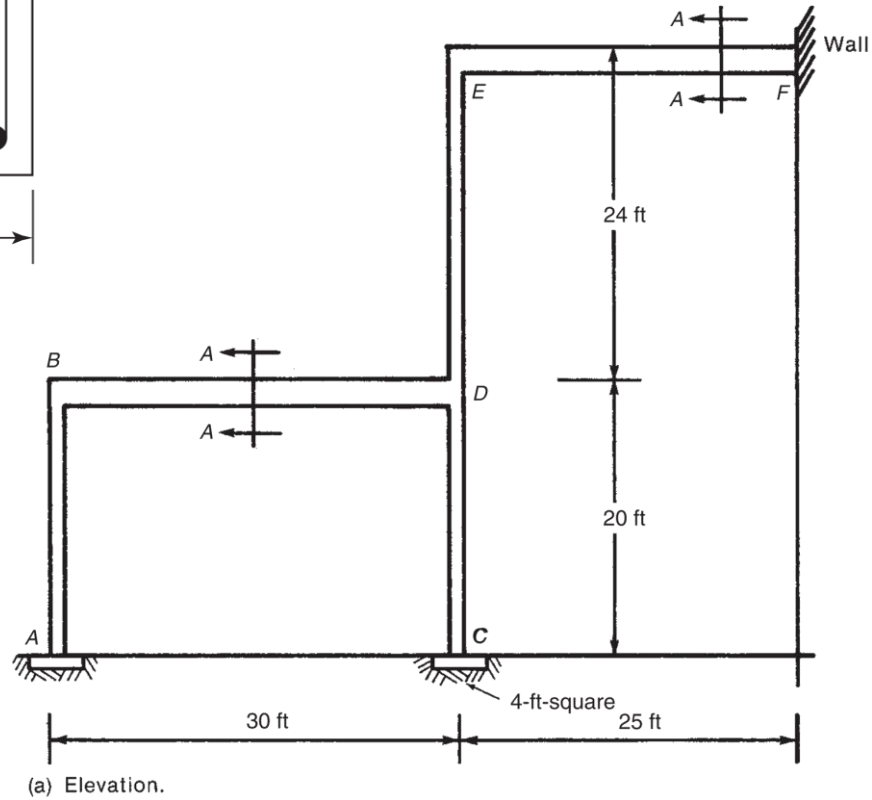
$$A_{st}(\text{req'd}) = 0.012 \times A_g = 0.012 \times (14 \text{ in.})^2 = 2.35 \text{ in.}^2$$

Use four No. 7 bars, $A_{st} = 4 \times 0.60 \text{ in.}^2 = 2.40 \text{ in.}^2$ In summary, **use a 14 in. × 14 in. column section with four No. 7 bars,** as shown in Fig. 12-29. ■

Solved Example

Design of a Slender Restrained Column in Nonsway (Braced) Frame





(a) Elevation.

Behavior and Design of Restrained Columns in Sway Frames

Nonsway and Sway Moments

M_{ns} and M_s Moments

Two different types of moments occur in frames:

1. Moments due to loads not causing appreciable sway, M_{ns}
2. Moments due to loads causing appreciable sway, M_s

The slenderness effects of these two kinds of moments are considered separately in the ACI Code design process because each is magnified differently as the individual columns deflect and as the entire frame deflects [12-22]. Column moments that cause no appreciable sway are magnified when the column deflects by an amount δ relative to its original straight axis such that the moments at points along the length of the column exceed those at the ends. In Section 12-2, this is referred to as the *member stability* effect or $P-\delta$ effect, where the lower case δ refers to deflections relative to the chord joining the ends of the column.

On the other hand, the column moments due to lateral loads can cause appreciable sway. They are magnified by the $P-\Delta$ moments resulting from the sway deflections, Δ, at joints in the frame, as indicated by Eq. (12-35). This is referred to as the $P-\Delta$ effect or *lateral drift* effect.

ACI Code Section 2.1 defines the *nonsway moment*, M_{ns}, as the factored end moment on a column due to loads that cause *no appreciable sidesway*, as computed by a first-order elastic frame analysis. These moments result from gravity loads. The 1977 through 1989 ACI Commentaries [12-13] defined "no appreciable sway" as being a lateral deflection of $\Delta/\ell \leq 1/1500$ of the story height at factored load levels. Gravity loads will cause small lateral deflections, except in the case of symmetrically loaded, symmetrical frames. The *sway moment*, M_s, is defined in ACI Code Section 2.1 as the factored end moment on a column due to loads which cause appreciable sidesway, calculated by a first-order elastic frame analysis. These moments result from either lateral loads or large unsymmetrical gravity loads, or gravity loads on highly unsymmetrical frames.

Nonsway and Sway Moments

Treating the $P-\delta$ and $P-\Delta$ moments separately simplifies design. The nonsway moments frequently result from a series of pattern loads, as was discussed in Chapter 10. The pattern loads can lead to a moment envelope for the nonsway moments. The maximum end moments from the moment envelope are then combined with the magnified sway moments from a second-order analysis or from a sway moment-magnifier analysis.

Computation of δ_s

Computation of $\delta_s M_s$

The following subsections discuss the ACI Code procedures for computing the magnified sway moments, $\delta_s M_s$, in a sway frame.

1. Computation of $\delta_s M_s$ Using Second-Order Analyses

ACI Code Sections 10.10.3 and 10.10.4 allow the use of second-order analyses to compute $\delta_s M_s$. Elastic second-order analyses were discussed in Section 12-6. If torsional displacements of the frame are significant, a three-dimensional second-order analysis should be used.

2. Computation of $\delta_s M_s$ Using Direct $P-\Delta$ Analysis

As part of the moment magnification method for sway frames, ACI Code Section 10.10.7 permits the use of a direct calculation of $P-\Delta$ moments via an equation similar to Eq. (12-42). In ACI Code Section 10.10.7, this is written as

$$\delta_s = \frac{1}{1 - Q} \geq 1 \qquad (12\text{-}43)$$

(ACI Eq. 10-20)

where

$$Q = \frac{\Sigma P_u \Delta_o}{V_{us} \ell_c} \qquad (12\text{-}21)$$

(ACI Eq. 10-10)

Although there is no easy way to incorporate torsional effects into the calculation of $\delta_s M_s$, Section 12-9 discusses the extension of the Q method to second-order torsion.

Computation of δ_s

3. Computation of $\delta_s M_s$ Using Sway-Frame Moment Magnifier

ACI Code Section 10.10.7 also allows the use of the traditional sway-frame moment magnifier for computing the magnified sway moments

$$\delta_s = \frac{1}{1 - \Sigma P_u/(0.75\ \Sigma P_c)} \geq 1 \qquad (12\text{-}44)$$

(ACI Eq. 10-21)

Here, ΣP_u and ΣP_c refer to the sums of the axial loads and critical buckling loads, respectively, for all the columns in the story being analyzed. In this case, the values of P_c are calculated by using the effective lengths, $k\ell_u$, evaluated for columns in a sway frame, with β_{ds} defined as

$$\beta_{ds} = \frac{\text{maximum factored sustained shear in the story}}{\text{total factored shear in the story}} \qquad (12\text{-}26b)$$

In most sway frames, the story shear is due to wind or seismic loads and hence is not sustained, resulting in $\beta_{ds} = 0$. The use of the summation terms in Eq. (12-44) accounts for the fact that sway instability involves all the columns and bracing members in the story. (See Fig. 12-30.) The format including the summations was first presented in [12-1].

If $(1 - \Sigma P_u/0.75\ \Sigma P_c)$ is negative, the load on the frame, ΣP_u, exceeds the buckling load for the story, ΣP_c, indicating that the frame is unstable. A stiffer frame is required.

Direct P-Δ Analysis for Sway Frames

Direct P–Δ Analysis for Sway Frames

The iterative calculation procedure described in the preceding section can be described mathematically as an infinite series. The sum of the terms in this series gives the second-order deflection

$$\Delta = \frac{\Delta_o}{1 - \gamma(\Sigma P_u)\Delta_o/(V_{us}\ell_c)} \tag{12-40}$$

where

V_{us} = shear in the story due to factored lateral loads acting on the frame above the story in question

ℓ_c = story height

ΣP_u = the total axial load in all the columns in the story

$\gamma \simeq 1.15$

Δ_o = first-order deflection of the top of a story relative to the bottom of the story, due to the story shear, V_{us}

Δ = second-order deflection

Both Δ_o and Δ refer to the lateral deflection of the top of the story relative to the bottom of the story.

Because the moments in the frame are directly proportional to the deflections, the second-order moments are

$$M = \delta_s M_s = \frac{M_o}{1 - \gamma(\Sigma P_u)\Delta_o/(V_{us}\ell_c)} \tag{12-41}$$

Direct P-Δ Analysis for Sway Frames

where M_o and M are the first- and second-order moments, respectively.

ACI Code Section 10.10.5.2 defines the stability index for a story as

$$Q = \frac{\Sigma P_u \Delta_o}{V_{us} \ell_c} \tag{12-21}$$

(ACI Eq. 10-10)

Substituting this into Eq. (12-41) and omitting the flexibility factor γ gives

$$\delta_s M_s = \frac{M_s}{1 - Q} \geq M_s \tag{12-42}$$

Reference [12-25] recommends that the use of Eq. (12-42) be limited to cases where δ_s is less than or equal to 1.5, because it becomes less accurate for higher values. This corresponds to $Q \leq 1/3$. For this reason, ACI Code Section 10.10.7.3 limits the use of Eq. (12-42) to $Q \leq 1/3$. Example 12-3 illustrates the impact of Eq. (12-42) on magnified moments in a frame.

Moments at the Ends of the Columns

Moments at the Ends of the Columns

The unmagnified nonsway moments, M_{ns}, are added to the magnified sway moments, $\delta_s M_s$, at each end of each column to get the total design moments:

$$M_1 = M_{1ns} + \delta_s M_{1s} \qquad \text{(12-45a)}$$

(ACI Eq. 10-18)

$$M_2 = M_{2ns} + \delta_s M_{2s} \qquad \text{(12-45b)}$$

(ACI Eq. 10-19)

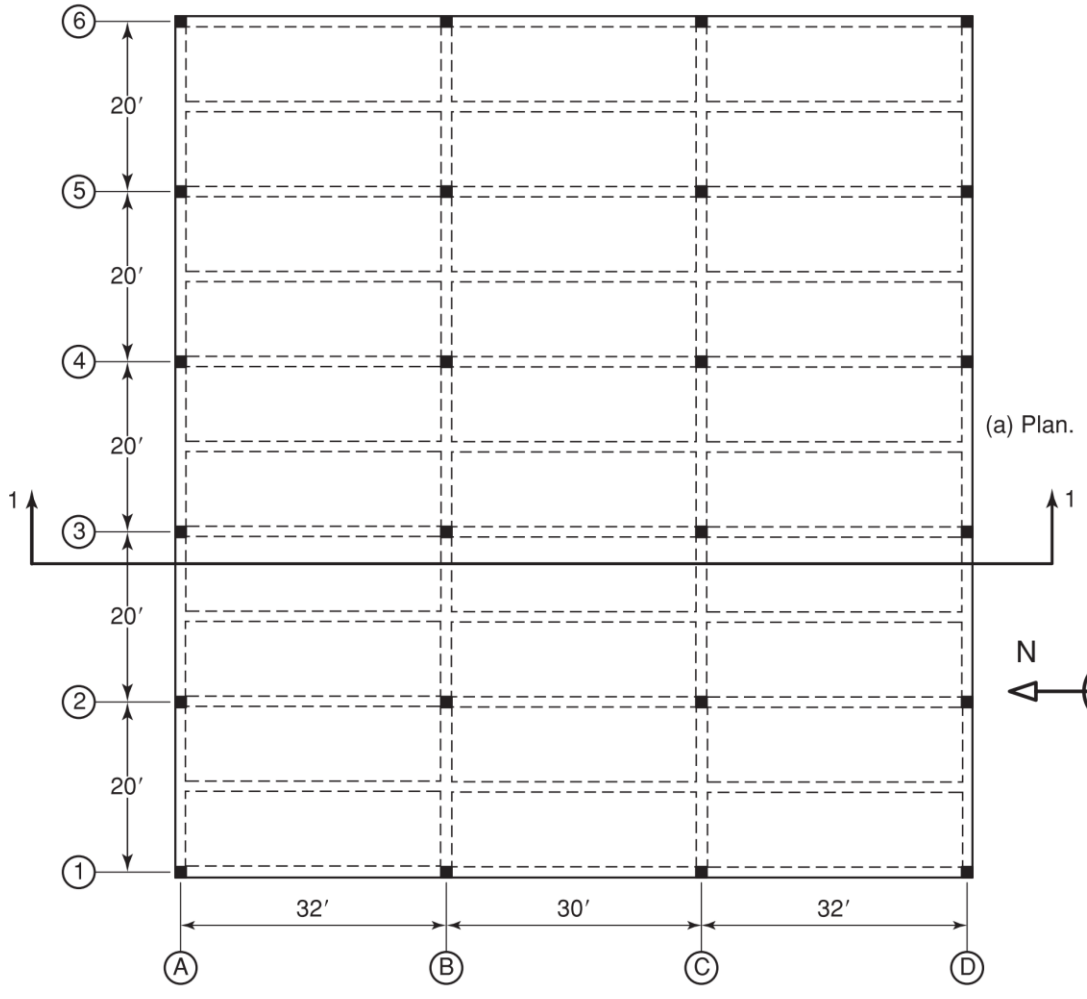
The addition is carried out for the moments at the top and bottom of each column. The larger absolute sum of the resulting end moments for a given column is called M_2, and the smaller is called M_1. By definition, M_2 is always taken as positive, and M_1 is taken as negative if the column is bent in double curvature.

Solved Example

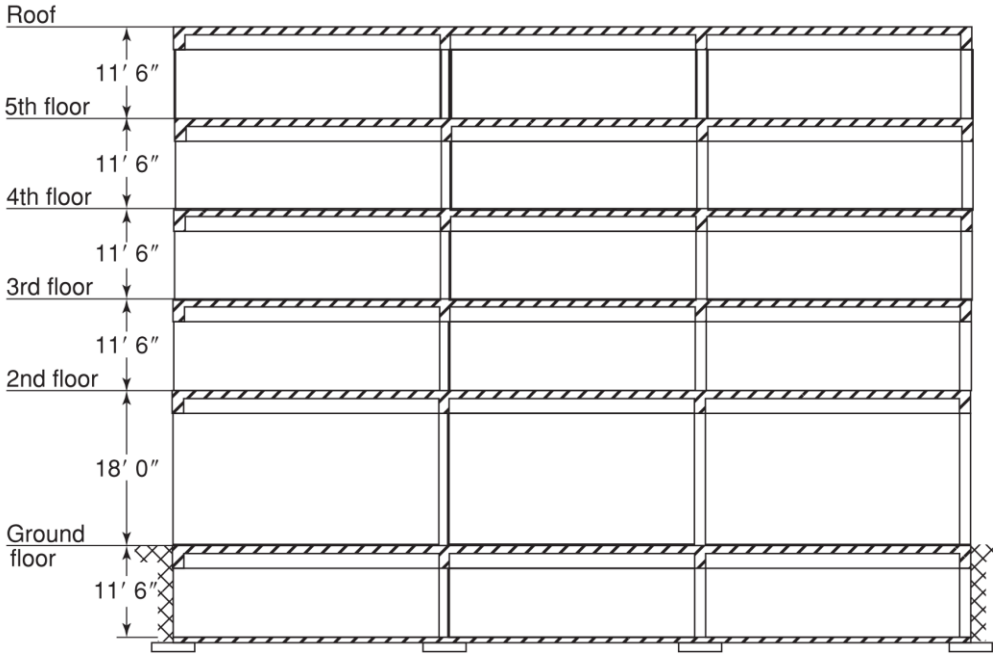
Design of a Slender Restrained Column in Sway Frame

Design of the Columns in a Sway Frame

Figure 12-32 shows the elevation and main floor plan for a five-story building. The building is clad with nonstructural precast panels. There are no structural walls or other bracing. The floor beams in the north–south direction are all 18 in. wide with an overall depth of 30 in. The floor slabs are 6 in. thick. Assume all the floors are to be designed for a superimposed dead load of 20 psf plus a live load of 80 psf, which includes a 20 psf partition loading. The roof is assumed to have similar-sized structural members, carries a superimposed dead load of 25 psf, and has a reduced live load, L_r, of 30 psf. Design an interior and exterior column in the first-story level for dead load, live load, and wind forces in the north–south direction. Assume we are using "service-level" wind forces, so the appropriate load factor is 1.6. Use $f'_c = 4000$ psi and $f_y = 60$ ksi.



(a) Plan.

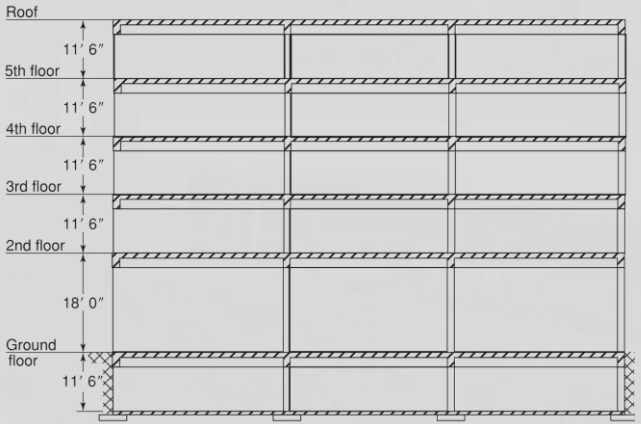


Slab thickness: 6″; Column size: 18″ × 18″; Beam size: 18″ × 30″

Section 1-1

Solved Example

Design of a Slender Restrained Column in Sway Frame



Slab thickness: 6″; Column size: 18″ × 18″; Beam size: 18″ × 30″

Assignment

Building Columns – A Summary

Buildings Columns

- Special Considerations
  - Limited size and shape due to architectural and space constraints
  - Generally, very high axial load, specially in lower floors of high-rise buildings
  - Consideration of differential axial shortening
  - Consideration of slenderness effects, specially in sway (unbraced) frames
  - Presence of biaxial moments in the corner columns due to gravity loads and all columns due to diagonal wind or seismic load direction
  - Use of high strength concrete and related special considerations
  - Requires high ductility in seismic zones

Slenderness procedure for Buildings

- **Step1.** Carry out frame analysis separately for all major load cases
  - Dead loads
  - Live loads
  - Wind loads
  - Seismic loads
- Step 2. Select a “Critical” floor
  - (maximum height, maximum loads, maximum deflection etc.)

Slenderness procedure for Buildings

- **Step 3.** Calculate “Factored Load” for various load combinations

  1) 1.4D +1.7L

  2) 1.05D +1.3L +1.3W

  3) 0.9D +1.3 W

Slenderness procedure for Buildings

- **Step 4.** For each load combination, Check sway conditions

$$Q = \frac{\sum P_U \Delta_0}{V_U l_C}$$

Equation 10-10
ACI 318-11

$$\sum P_U = P_{U1} + P_{U2} + P_{U3} \ldots\ldots$$

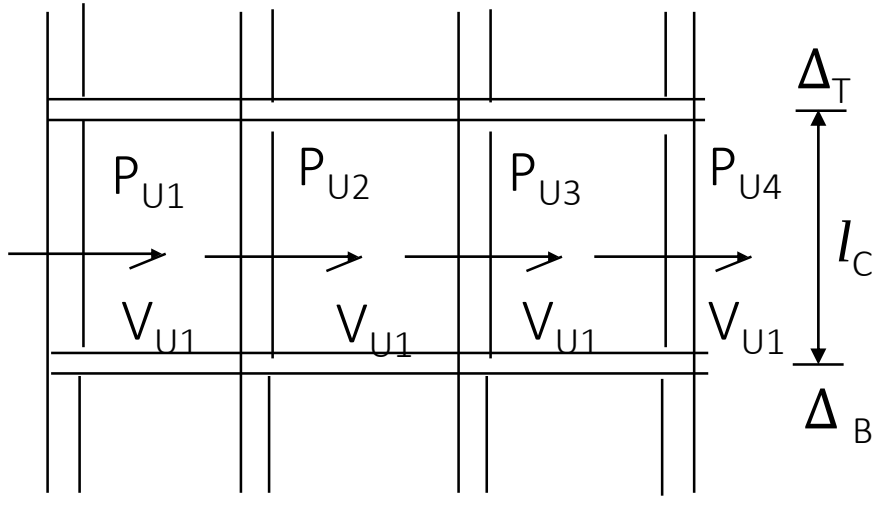
$$\Delta_0 = \Delta_T - \Delta_B$$

$$V_U = V_{U1} + V_{U2} + V_{U3} \ldots\ldots$$

$$l_C = Clear\ storey\ height\ (average)$$

$$If\ Q \le 0.05 : Non-sway case$$

$$Q > 0.05 : Sway Case$$



Slenderness procedure for Buildings

- **Step 5.** Determine Magnified Moment for each load combination
  - If combination is **non-sway,** then $M_m = M\,\delta_{ns}$
  - If combination is "**Sway**" then $M_m = M_{ns}\,\delta_{ns} + M_s\,\delta_s$
  - Usually, 1.05D +1.3L+1.3W

Non-sway part of combination 1.05D + 1.3L

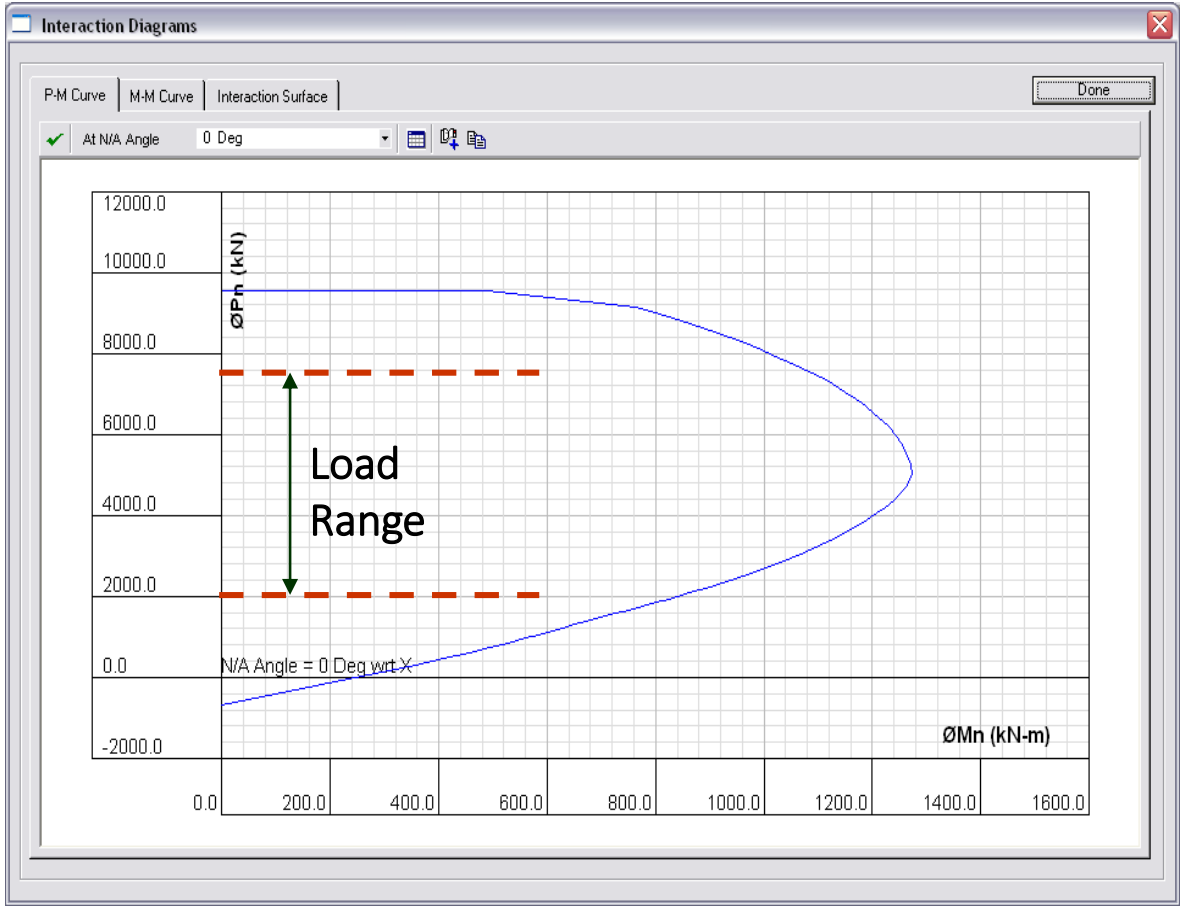
Sway part of combination 1.3W

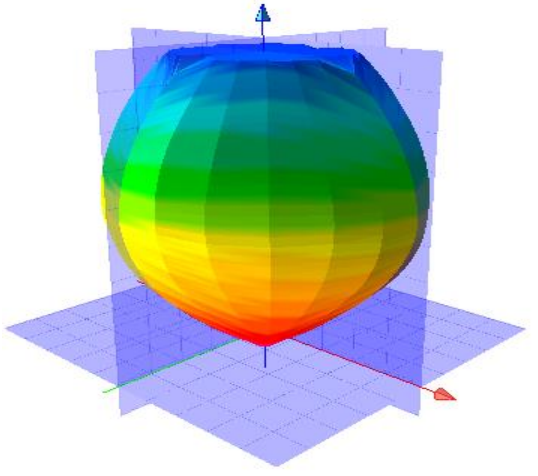
The Factors Affecting Moment Magnification

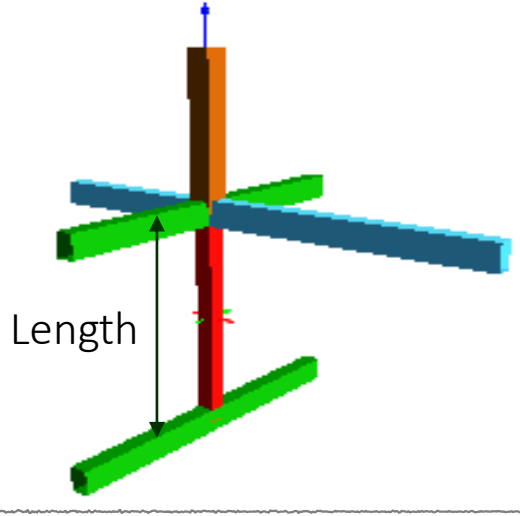
Parametric Study

- Computation of Slenderness Effects for 3 column sections for different axial load and lengths.
  - A = 30x30 cm
  - B = 40x40 cm
  - C = 80x80 cm
- Braced (Non-Sway) frames assuming shear walls prevent large lateral displacements.

Column Section Shape and Properties

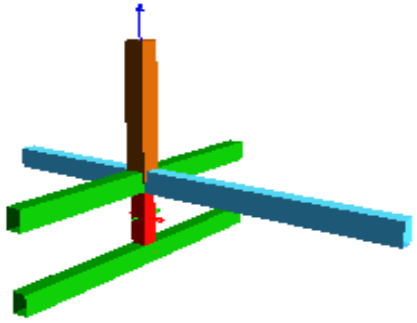




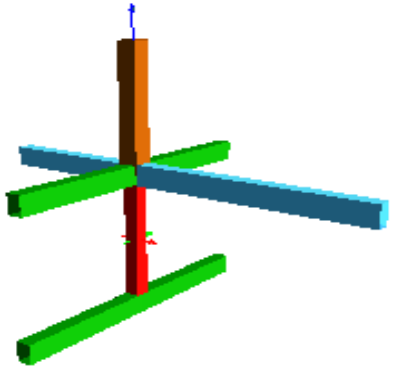


A30-Bracing Conditions

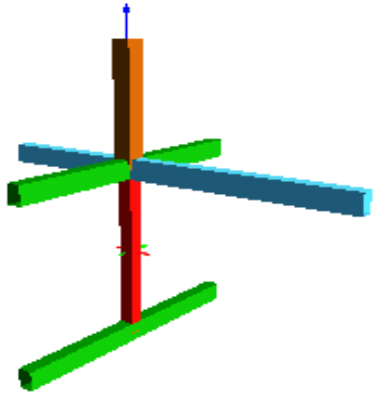
- Column Cross-Section = 30cmx30cm reinforced with 6-d20
- Connecting Members
  - Beam on Right:
    - Length = 5 m
    - Cross-section = 30cmx50cm
  - Beam on Left:
    - Length = 3 m
    - Cross-section = 30cmx50cm
  - Column Above
    - Length = 3m
    - Cross-section = 40cmx40cm
- Fixed at Base
- The column is part of a non-sway structure



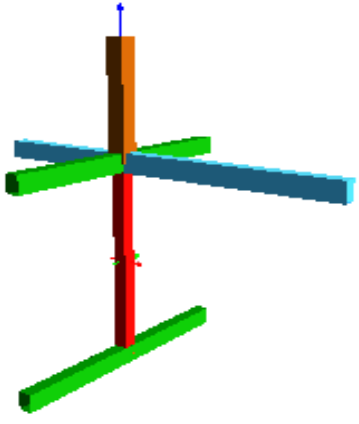
kl/r=14.5



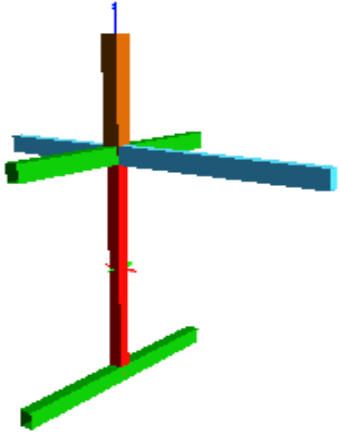
kl/r=28.9



kl/r=38.1

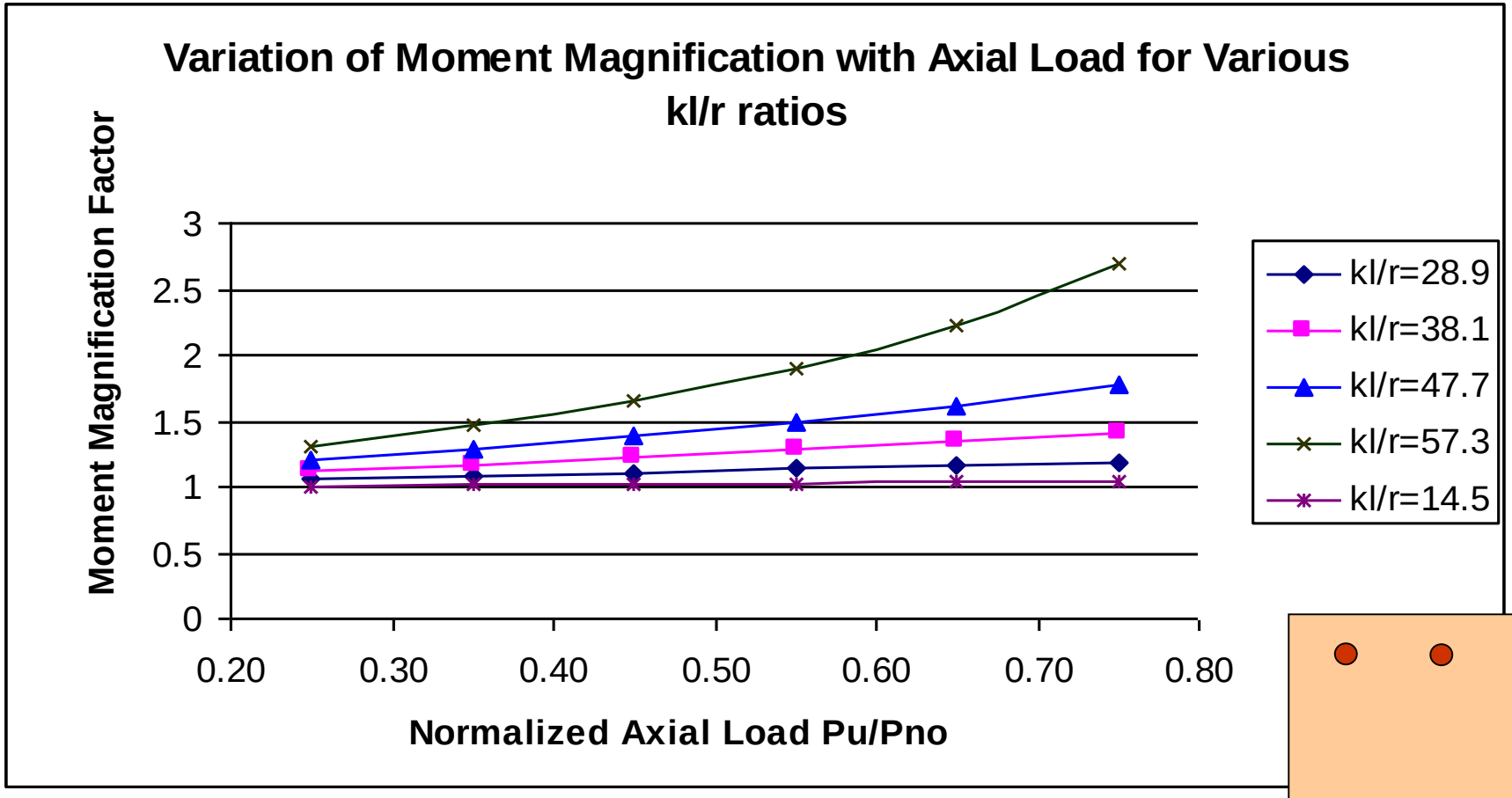


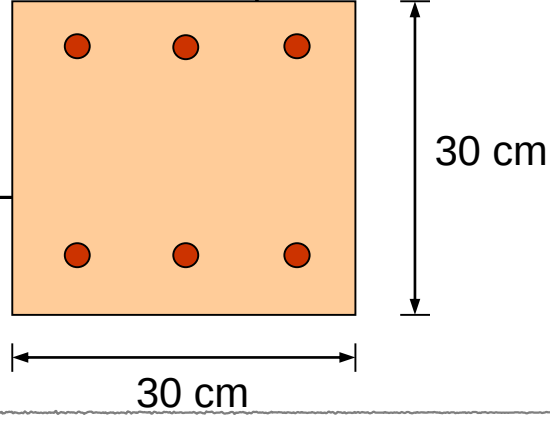
kl/r=47.7



kl/r=57.3

A30 - Variation in kl/r

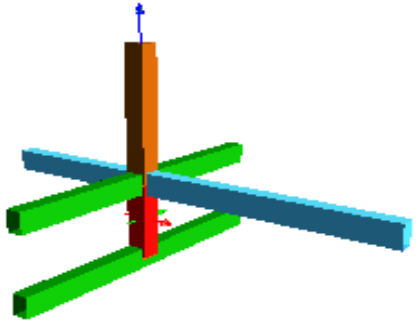




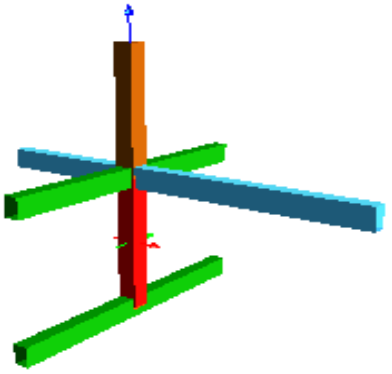
A30 – Moment Magnification

B40 - Bracing Conditions

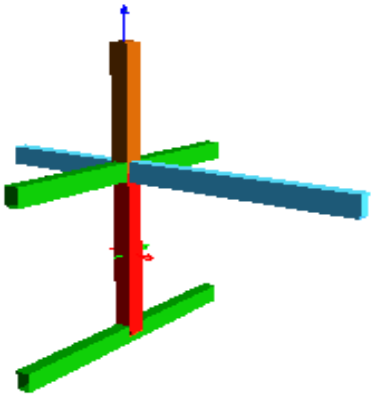
- Column Cross-Section = 40cmx40cm reinforced with 6-d20
- Connecting Members
  - Beam on Right:
    - Length = 5 m
    - Cross-section = 30cmx50cm
  - Beam on Left:
    - Length = 3 m
    - Cross-section = 30cmx50cm
  - Column Above
    - Length = 3m
    - Cross-section = 40cmx40cm
- Fixed at Base
- The column is part of a non-sway structure



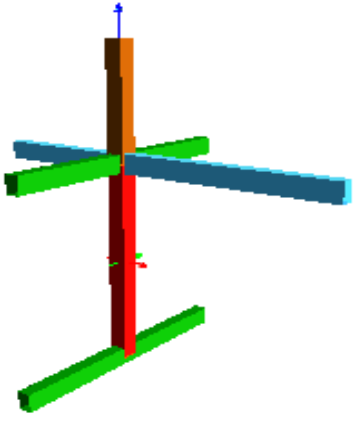
kl/r=11



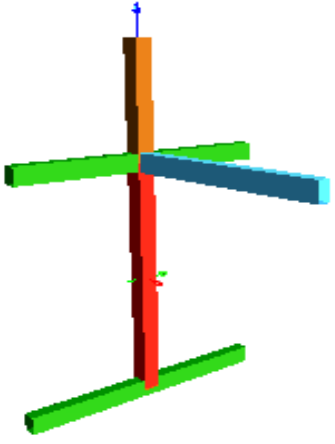
kl/r=22



kl/r=29

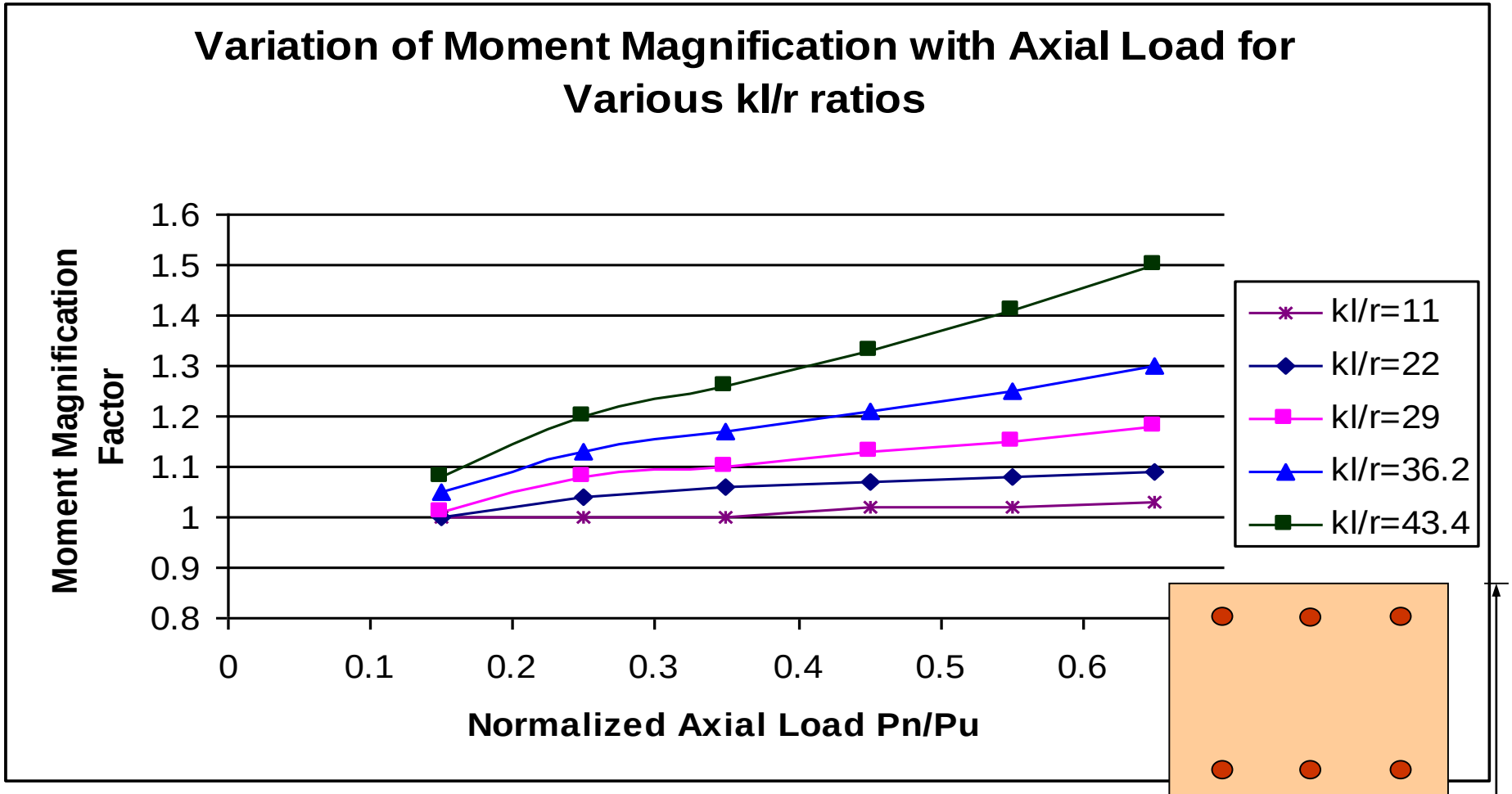


kl/r=36.2



kl/r=43.4

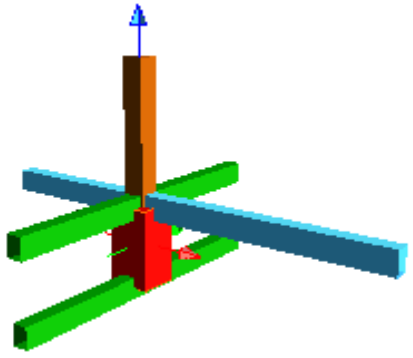
B40 - Variation in kl/r



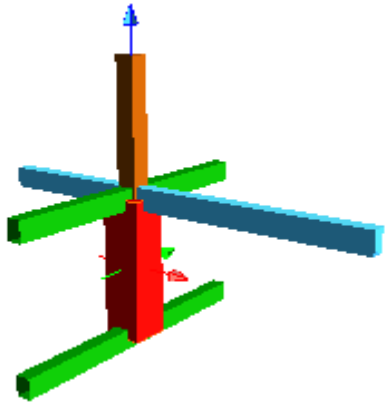
B40 – Moment Magnification

C80 - Bracing Conditions

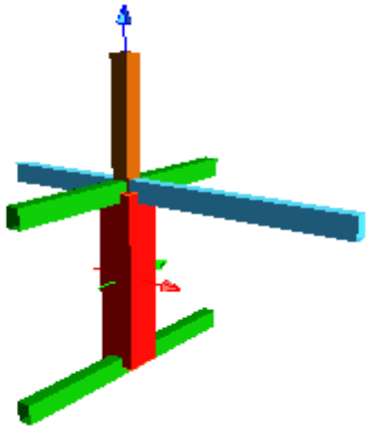
- Column Cross-Section = 80cmx80cm reinforced with 6-d20
- Connecting Members
  - Beam on Right:
    - Length = 5 m
    - Cross-section = 30cmx50cm
  - Beam on Left:
    - Length = 3 m
    - Cross-section = 30cmx50cm
  - Column Above
    - Length = 3m
    - Cross-section = 40cmx40cm
- Fixed at Base
- The column is part of a non-sway structure



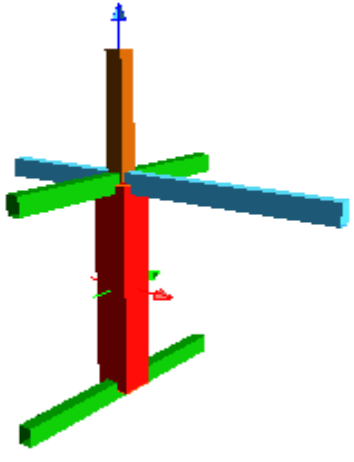
kl/r=5.5



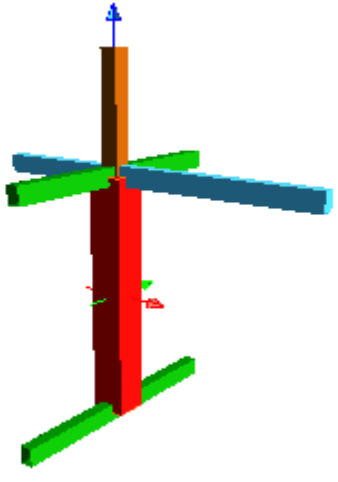
kl/r=11.2



kl/r=14.9

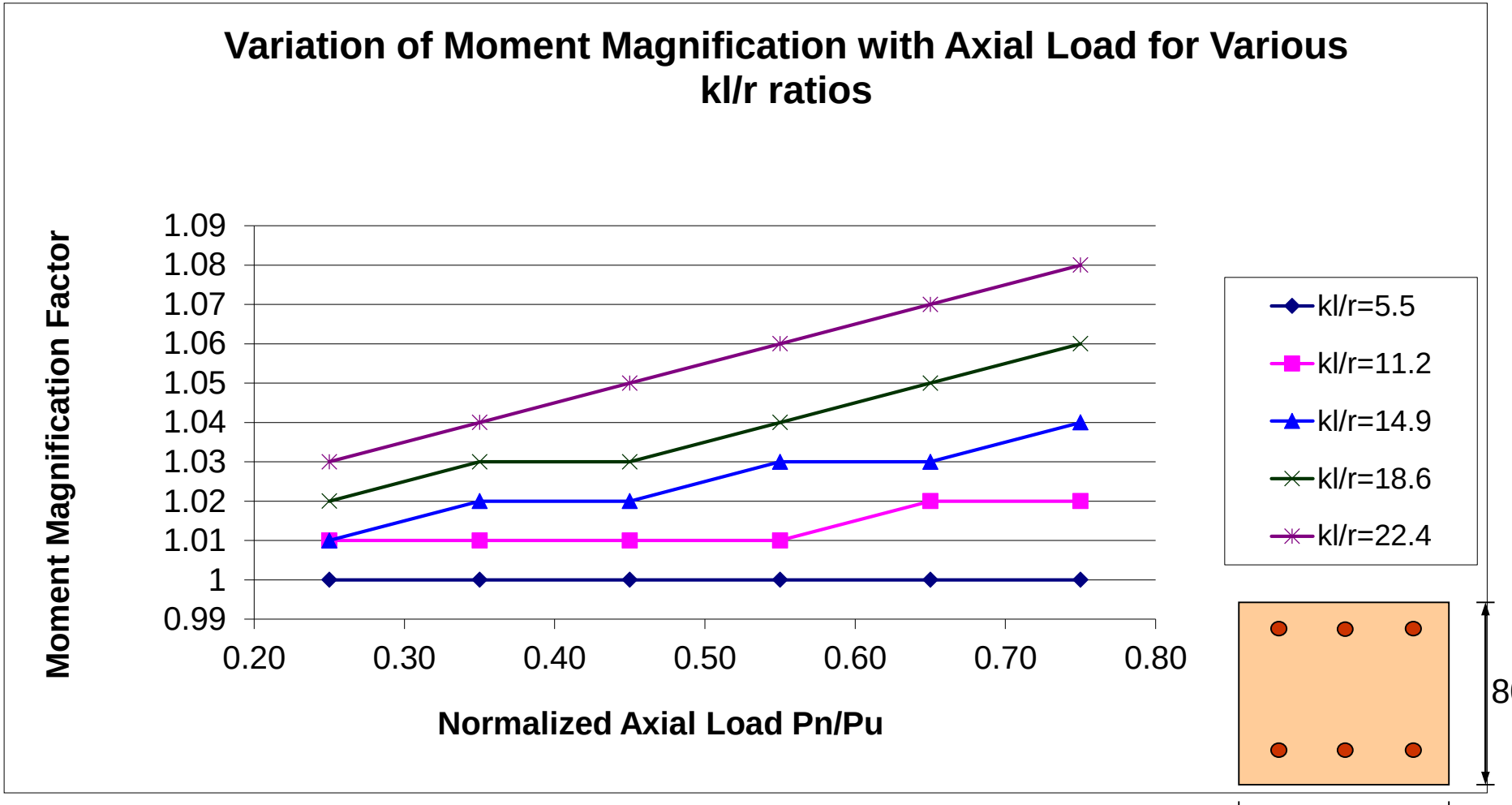


kl/r=18.6



kl/r=22.4

C80 - Variation in kl/r



C80 – Moment Magnification

Ductility of Column Cross-sections

Ductility of Unconfined Column Sections

- The curvature of the section is influenced by the axial load, hence there is no unique M-φ relationship for a given column section.
- However, it is possible to plot the combination of axial load P and Moment M which cause the section to reach the ultimate capacity.
- It is evident that the ductility of the column section is significantly reduced by the presence of axial load.
- The axial load levels greater than the balanced failure load, the ductility decreases, being due only to the inelastic deformation of the concrete.

Ductility of Unconfined Column Sections

- At the levels of load less than the balance load, the ductility increases as the load level is reduced.
- Because of the brittle failure of the unconfined columns at moderate axial load, ACI code recommends that the ends of the columns in ductile frame in earthquake areas be confined by closely spaced transverse reinforcement when axial load is greater than 0.4 times balanced load.

Ductility of Confined Column Sections

- The concrete section may fail in brittle manner if there is not enough confinement to the concrete.
- If the compression zone of a member is confined by closely spaced transverse reinforcement in the form of stirrups, ties , hoops or spirals, the ductility of the concrete may increased significantly.
- When compressive stress approaching the compressive strength of concrete, the transverse strains in the concrete increased rapidly and the concrete expands against the transverse reinforcement.

Ductility of Confined Column Sections

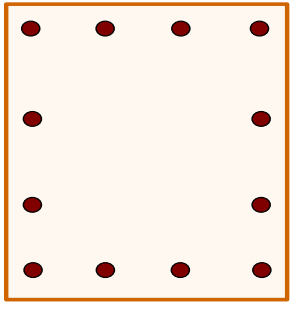
- The retaining pressure applied by the reinforcement to the concrete considerably improves the stress-strain behavior of the concrete at higher strain. Thus, helps to improve the ductility of the member.
- Circular spirals confine the concrete more effectively than rectangular stirrups, ties or hoops because confining steel in the shape of circle applies a uniform radial pressure to the concrete, whereas a rectangle tends to confine the concrete mainly at the corner.

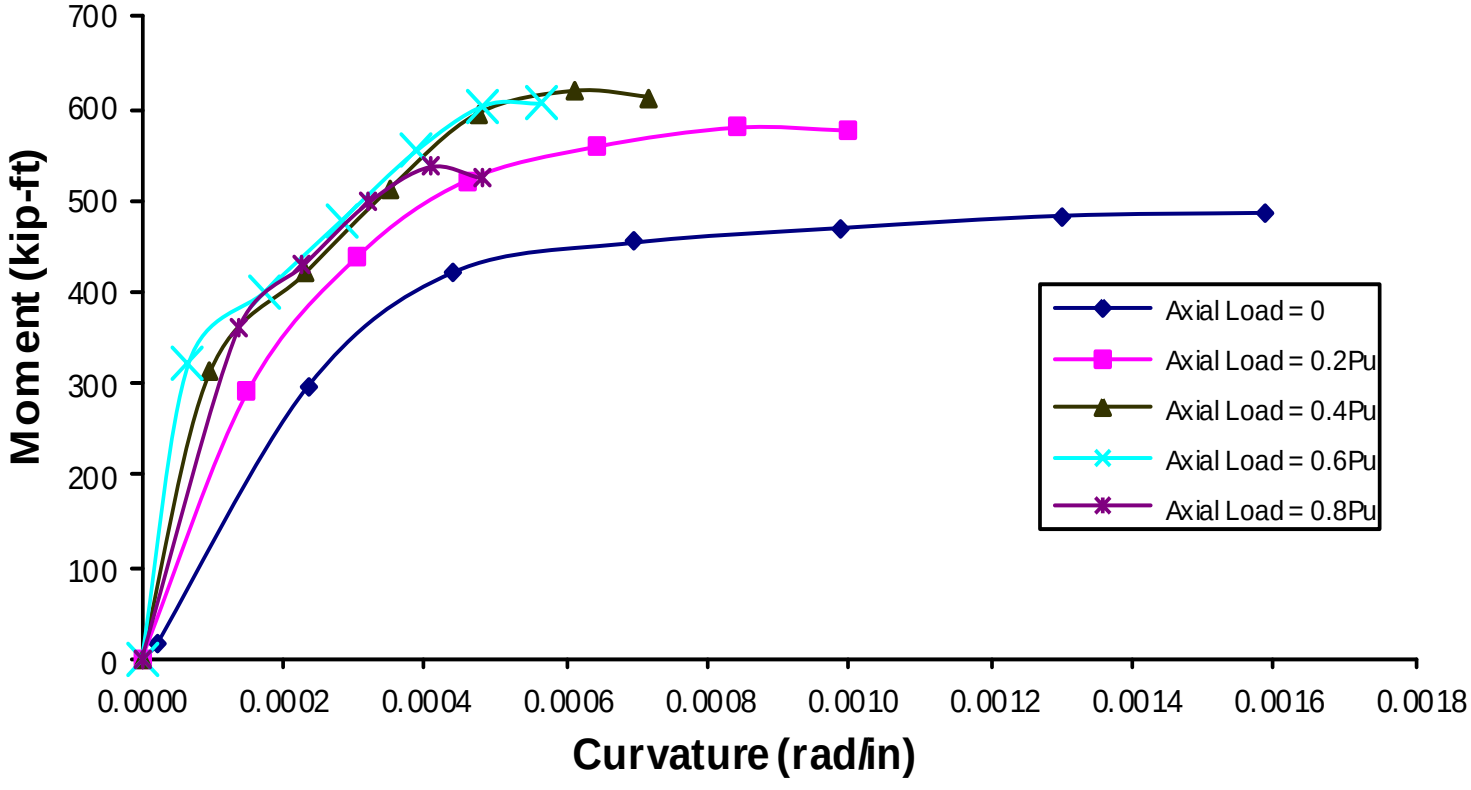
M-ϕ Relationship and Ductility

- Effect of Axial Load
- Effect of Reinforcement ratio
- Effect of Compression Steel
- Effect of Confinement Model
- Effect of Confinement Shape

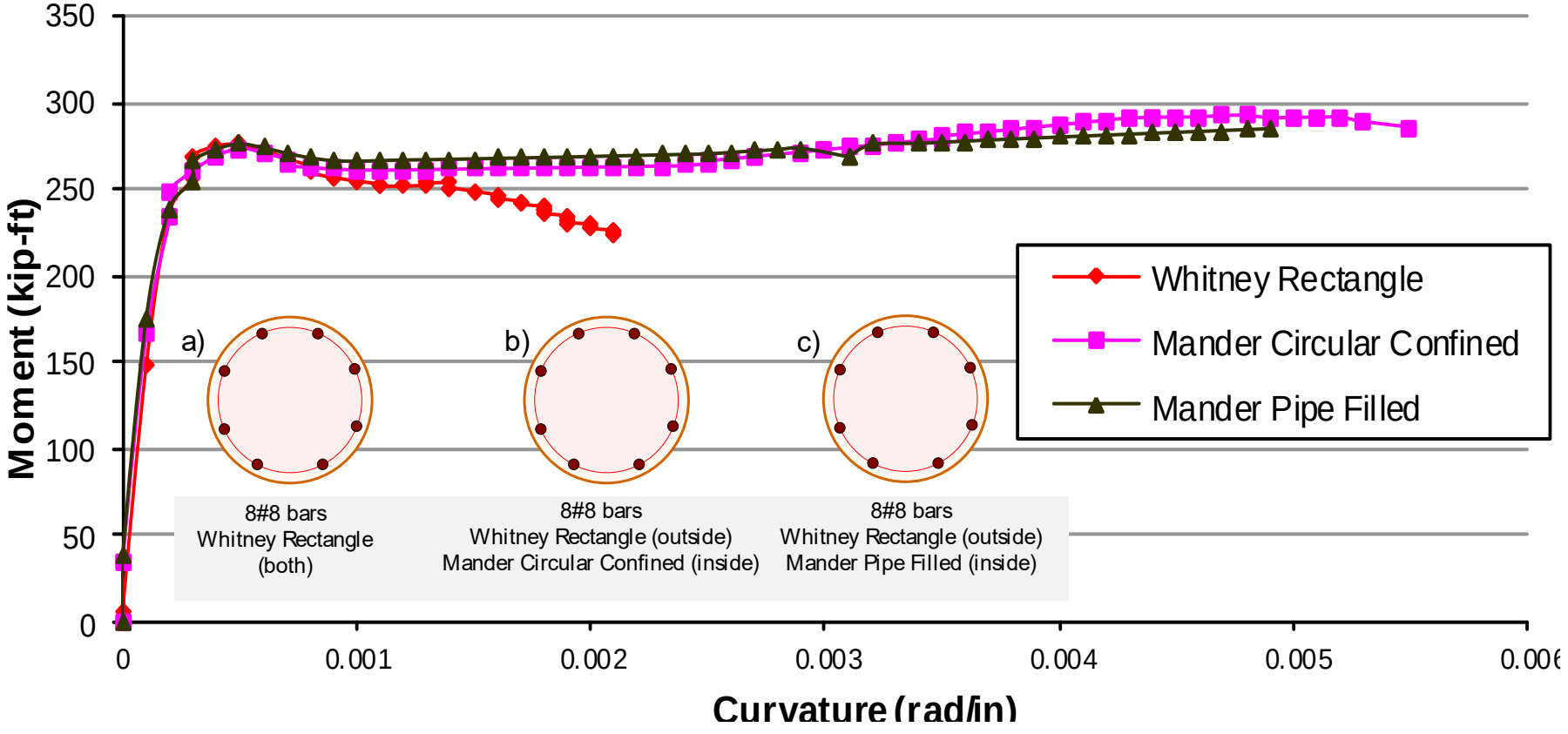
Axial Load and Ductility

Very Important !!!

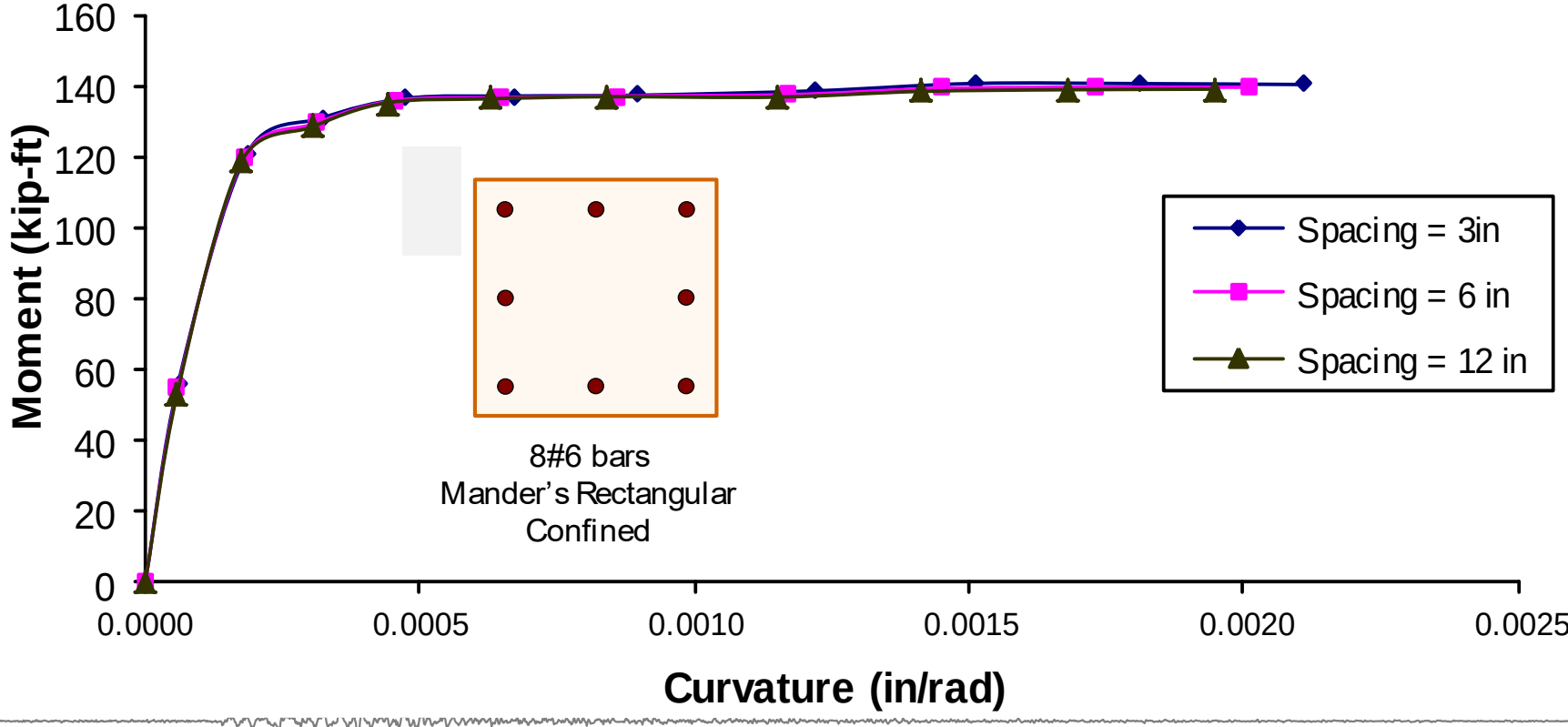




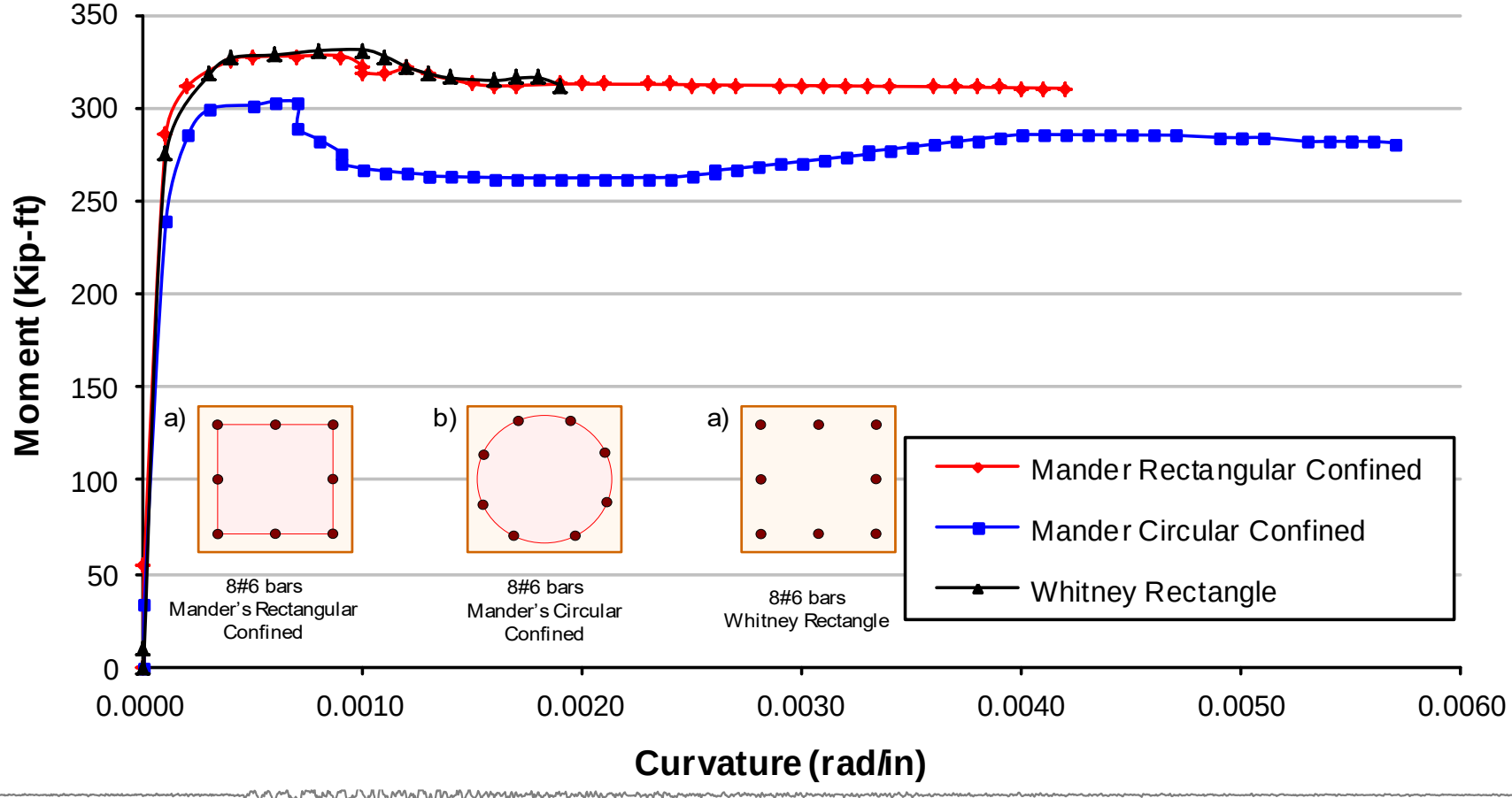
Effect of Confinement



Confinement Steel and Ductility



Confinement Shape and Ductility



Strategies to Improve Ductility

- Use low flexural reinforcement ratio
- Add compression reinforcement
- Add confining reinforcement

P-Δ Analysis in SAP 2000 and ETABS 2016/17

- The program can include the P-Δ effects in almost all Non-linear analysis types.
- Specific P-Δ analysis can also be carried out.
- The P-Δ analysis basically considers the geometric nonlinear effects directly.
- The material nonlinear effects can be handled by modification of cross-section properties.
- The Buckling Analysis is not the same as P- Δ Analysis.
- No magnification of moments is needed if P- Δ Analysis has been carried out.


Thank You!